BCube Prompt Engine v3.0

Complete Production Standard

Founder Edition - Version 3.0.0

BCube Prompt Engine v3.0 is the authoritative production system used to create deterministic, traceable, child-first, teacher-enabled, parent-connected, print-ready BCube Gold Production Prompts.

> *Single source of truth: this Markdown document and its linked module files govern the generated Word and PDF editions.*

Document control

Field	Value
Product	BCube Prompt Engine v3.0
Edition	Founder Edition
Version	3.0.0
Status	Active
Owner	BCube Future Academy
Release date	15 July 2026
Change model	Semantic versioning and governed approval

Master production overview

Purpose

BCube Prompt Engine™ v3.0 is the production specification used to generate every BCube illustration prompt. It combines educational intent, publishing rules, visual standards, teacher guidance, AI generation rules, and quality validation into one reusable prompt engine.

1. Engine Architecture

Objective: Standardize this layer of the prompt engine across all BCube publications.

- Master Prompt
- Reusable Modules
- Page Data Schema
- Prompt Compiler
- Validation Engine

Implementation: Every production prompt is assembled from reusable modules plus page-specific data.

Acceptance: Prompts compile successfully and satisfy all validation checks before illustration generation.

2. Universal Modules

Objective: Standardize this layer of the prompt engine across all BCube publications.

- Publishing Module
- Design Module
- Visual Grammar Module
- Character Module
- Educational Module
- Teacher Module
- Parent Module
- Story Module
- QA Module
- Print Module

Implementation: Every production prompt is assembled from reusable modules plus page-specific data.

Acceptance: Prompts compile successfully and satisfy all validation checks before illustration generation.

3. Page Data Schema

Objective: Standardize this layer of the prompt engine across all BCube publications.

- Book
- Series
- Age
- Page Number
- Page Title
- Learning Objective
- Competency
- Activity
- Illustration Focus
- Assessment Evidence

Implementation: Every production prompt is assembled from reusable modules plus page-specific data.

Acceptance: Prompts compile successfully and satisfy all validation checks before illustration generation.

4. Prompt Assembly Rules

Objective: Standardize this layer of the prompt engine across all BCube publications.

- Load universal modules
- Insert page data
- Apply book theme

- Run validation
- Generate final prompt

Implementation: Every production prompt is assembled from reusable modules plus page-specific data.

Acceptance: Prompts compile successfully and satisfy all validation checks before illustration generation.

5. Publishing Specification

Objective: Standardize this layer of the prompt engine across all BCube publications.

- A4 portrait
- 300 DPI
- CMYK
- Safe margins
- Print-ready layout
- Brand compliance

Implementation: Every production prompt is assembled from reusable modules plus page-specific data.

Acceptance: Prompts compile successfully and satisfy all validation checks before illustration generation.

6. Educational Specification

Objective: Standardize this layer of the prompt engine across all BCube publications.

- Learning objective
- Future skill
- Child outcome
- Teacher interaction
- Parent extension

Implementation: Every production prompt is assembled from reusable modules plus page-specific data.

Acceptance: Prompts compile successfully and satisfy all validation checks before illustration generation.

7. Illustration Specification

Objective: Standardize this layer of the prompt engine across all BCube publications.

- Star mascot
- Teacher
- Children
- Environment
- Composition
- Colour harmony
- Accessibility

Implementation: Every production prompt is assembled from reusable modules plus page-specific data.

Acceptance: Prompts compile successfully and satisfy all validation checks before illustration generation.

8. AI Generation Rules

Objective: Standardize this layer of the prompt engine across all BCube publications.

- Consistent terminology
- Structured outputs
- Negative instructions
- No ambiguity
- Reusable variables

Implementation: Every production prompt is assembled from reusable modules plus page-specific data.

Acceptance: Prompts compile successfully and satisfy all validation checks before illustration generation.

9. Quality Validation Engine

Objective: Standardize this layer of the prompt engine across all BCube publications.

- Educational QA
- Visual QA
- Publishing QA
- Accessibility QA
- Brand QA
- Final Gold Score

Implementation: Every production prompt is assembled from reusable modules plus page-specific data.

Acceptance: Prompts compile successfully and satisfy all validation checks before illustration generation.

10. Version Governance

Objective: Standardize this layer of the prompt engine across all BCube publications.

- Semantic versioning
- Module versioning
- Change log
- Review cycle
- Approval workflow

Implementation: Every production prompt is assembled from reusable modules plus page-specific data.

Acceptance: Prompts compile successfully and satisfy all validation checks before illustration generation.

Master Prompt Flow

Curriculum → Page Data → BCube Prompt Engine™ → AI Generation → Quality Validation → Publishing Approval → Final Book

Initial Deliverables

- Master Prompt Template - Universal Module Library - Page Data Schema - Prompt Compiler Rules - Validation Checklist - Gold Production Prompt Standard™ Integration

Universal module library

Overview

This document defines the reusable modules that are combined with page-specific data to produce every BCube Gold Production Prompt™. These modules should not be duplicated across prompts; they are referenced by version and assembled by the Prompt Engine.

Module M01 – Publishing

Print specifications, A4 portrait, 300 DPI, CMYK, bleed, safe margins, whitespace, trim safety, export readiness.

Mandatory Elements

- A4 portrait
- 300 DPI
- CMYK
- 5 mm bleed
- 10 mm safe margin
- Print-ready composition

Module Inputs

Consumes standardized page metadata and global BCube standards.

Module Outputs

Produces reusable prompt content for compilation into the final production prompt.

Acceptance Criteria

Module output is internally consistent, reusable across books, and conforms to BCube EduOS™.

Module M02 – Design Language

Layout grids, typography zones, title placement, activity area, instruction hierarchy and page balance.

Mandatory Elements

- Grid
- Typography
- Hierarchy

- White space
- Layout balance
- Accessibility

Module Inputs

Consumes standardized page metadata and global BCube standards.

Module Outputs

Produces reusable prompt content for compilation into the final production prompt.

Acceptance Criteria

Module output is internally consistent, reusable across books, and conforms to BCube EduOS™.

Module M03 – Visual Grammar

Composition rules that guide eye movement and educational clarity.

Mandatory Elements

- Primary focal point
- Secondary focal point
- Eye flow
- Contrast
- Visual rhythm
- Minimal clutter

Module Inputs

Consumes standardized page metadata and global BCube standards.

Module Outputs

Produces reusable prompt content for compilation into the final production prompt.

Acceptance Criteria

Module output is internally consistent, reusable across books, and conforms to BCube EduOS™.

Module M04 – Character System

Defines Star, teachers, children and recurring characters.

Mandatory Elements

- Star mascot
- Teacher behaviour
- Child diversity
- Expressions
- Positive body language
- Consistency

Module Inputs

Consumes standardized page metadata and global BCube standards.

Module Outputs

Produces reusable prompt content for compilation into the final production prompt.

Acceptance Criteria

Module output is internally consistent, reusable across books, and conforms to BCube EduOS™.

Module M05 – Educational Engine

Educational metadata inserted into every prompt.

Mandatory Elements

- Learning objective
- Competency
- Future skill
- Expected outcome
- Assessment evidence

Module Inputs

Consumes standardized page metadata and global BCube standards.

Module Outputs

Produces reusable prompt content for compilation into the final production prompt.

Acceptance Criteria

Module output is internally consistent, reusable across books, and conforms to BCube EduOS™.

Module M06 – Teacher Engine

Teacher actions embedded in illustrations.

Mandatory Elements

- Facilitation
- Questioning
- Scaffolding
- Encouragement
- Observation

Module Inputs

Consumes standardized page metadata and global BCube standards.

Module Outputs

Produces reusable prompt content for compilation into the final production prompt.

Acceptance Criteria

Module output is internally consistent, reusable across books, and conforms to BCube EduOS™.

Module M07 – Parent Engine

Parent extension activities and home learning cues.

Mandatory Elements

- Conversation prompt
- Home activity
- Positive reinforcement
- Extension idea

Module Inputs

Consumes standardized page metadata and global BCube standards.

Module Outputs

Produces reusable prompt content for compilation into the final production prompt.

Acceptance Criteria

Module output is internally consistent, reusable across books, and conforms to BCube EduOS™.

Module M08 – Story Engine

Every illustration tells a meaningful micro-story.

Mandatory Elements

- Beginning
- Interaction
- Learning moment
- Positive ending

Module Inputs

Consumes standardized page metadata and global BCube standards.

Module Outputs

Produces reusable prompt content for compilation into the final production prompt.

Acceptance Criteria

Module output is internally consistent, reusable across books, and conforms to BCube EduOS™.

Module M09 – Illustration Engine

Scene composition and visual execution.

Mandatory Elements

- Perspective
- Lighting
- Colour harmony
- Environment
- Props

- Accessibility

Module Inputs

Consumes standardized page metadata and global BCube standards.

Module Outputs

Produces reusable prompt content for compilation into the final production prompt.

Acceptance Criteria

Module output is internally consistent, reusable across books, and conforms to BCube EduOS™.

Module M10 – Quality Engine

Validation before publishing.

Mandatory Elements

- Educational QA
- Visual QA
- Brand QA
- Accessibility QA
- Print QA
- Gold score

Module Inputs

Consumes standardized page metadata and global BCube standards.

Module Outputs

Produces reusable prompt content for compilation into the final production prompt.

Acceptance Criteria

Module output is internally consistent, reusable across books, and conforms to BCube EduOS™.

Prompt Compilation Sequence

M01 Publishing → M02 Design → M03 Visual Grammar → M04 Character → M05 Educational → M06 Teacher → M07 Parent → M08 Story → M09 Illustration → M10 Quality → Page-specific Data → Final BCube Gold Production Prompt™

Foundation and Normative Language

Foundation and Normative Language

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Preface

This document is the beginning of the master BCube Prompt Engine™ v3.0 Complete Production Standard. It is intended to grow into the single authoritative reference governing every BCube educational prompt, illustration, workbook page, publishing workflow and quality process.

Chapter 1 – Normative Language

- MUST: Mandatory requirement that every production prompt shall satisfy.
- SHALL: Equivalent mandatory requirement used for formal specifications.
- SHOULD: Recommended practice unless a justified exception exists.
- MAY: Optional capability.
- MUST NOT: Prohibited behaviour.

Chapter 2 – Rule Identification

- PE-: Publishing Engine
- DE-: Design Engine
- VG-: Visual Grammar Engine
- EE-: Educational Engine
- TE-: Teaching Engine
- PP-: Parent Partnership Engine
- IE-: Illustration Engine
- CE-: Character Engine
- QA-: Quality Assurance Engine
- PC-: Prompt Compiler

Chapter 3 – Standard Rule Template

- Rule ID: Unique identifier
- Requirement: Normative statement
- Rationale: Why the rule exists
- Inputs: Required metadata
- Outputs: Generated specification
- Validation: How compliance is verified
- Exceptions: Approved deviations
- Cross References: Linked rules

Chapter 4 – Master Prompt Data Model

- Global Metadata: Publisher, version, language
- Book Metadata: Series, grade, curriculum

- Unit Metadata: Theme, competency
- Page Metadata: Title, objective, activity
- Compilation Metadata: Engine versions, QA status

Master Prompt Skeleton

1. Compiler Metadata 2. Publishing Engine Output 3. Design Engine Output 4. Visual Grammar Output 5. Educational Output 6. Teaching Output 7. Parent Partnership Output 8. Illustration Output 9. Character Output 10. QA Output 11. Final Page-Specific Instructions 12. Negative Constraints 13. Validation Summary

Expansion Roadmap

The master standard will continue by expanding every rule into detailed operational specifications, including rule identifiers, examples, non-examples, prompt fragments, validation logic, reusable templates, and compiler-ready outputs until the document becomes the definitive BCube production reference.

Phase 1 - Publishing Engine

Phase 1 - Publishing Engine

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

BCube Prompt Engine™ v3.0

Phase 1 — Publishing Engine™

Final Production Specification • Founder Edition • Version 3.0

The authoritative publishing and print-production rules used by the BCube Prompt Engine™ to compile consistent, child-friendly, premium, print-ready workbook page prompts.

Document Control

Field	Value
Document	BCube Prompt Engine™ v3.0 — Publishing Engine™
Document ID	BCPE-V3-PH1-PE
Status	FINAL
Owner	BCube Future Academy
Applies To	Nursery, LKG, UKG, Grade 1+ publishing prompts
Review Cycle	Annual or upon major production-system change
Change Control	BCube EduOS™ governance and semantic versioning

1. Purpose and Scope

The Publishing Engine™ is the first mandatory engine in the BCube Prompt Engine™ v3.0 compilation sequence. It converts broad creative intent into a precise publishing specification. Its purpose is to ensure that every generated page is physically producible, visually legible, developmentally appropriate, brand-consistent, and safe for commercial printing.

- Applies to covers, front matter, activity pages, review pages, certificates, teacher guidance pages and back covers.
- Controls page geometry, trim safety, bleed, live area, visual-to-text ratio, typography zones, print colour, image resolution and export readiness.
- Provides machine-readable rules that downstream Design, Visual Grammar, Illustration and QA engines inherit.
- Prevents common failures such as cropped content, tiny text, crowded activity zones, inconsistent page numbering, low-resolution artwork and incorrect logo placement.

2. Engine Position in the Compilation Pipeline

The Publishing Engine™ runs before all other visual engines because page geometry and production limits constrain every later decision.

Stage	Engine	Primary Output
1	Publishing Engine™	Physical page and print specification
2	Design Engine™	Layout hierarchy and component placement
3	Visual Grammar Engine™	Focal point, eye flow and visual rhythm
4	Educational Engine™	Learning objective and evidence
5	Teaching Engine™	Teacher-child interaction
6	Parent Partnership Engine™	Home extension
7	Illustration Engine™	Scene execution
8	Character Engine™	Character consistency
9	Quality Assurance Engine™	Validation and scoring
10	Prompt Compiler™	Final production prompt

3. Mandatory Page Geometry

Unless a product brief explicitly authorizes another format, all standard BCube workbook pages use A4 portrait geometry.

Parameter	Standard	Tolerance	Rule
Trim size	210 × 297 mm	±0.5 mm	Final printed page size
Orientation	Portrait	None	Landscape requires product-level approval
Bleed	3 mm minimum	3–5 mm	Background artwork extends through bleed
Safe margin	10 mm	Minimum 8 mm	No essential text or faces outside this zone
Binding margin	12 mm inside edge	12–15 mm	Increase for perfect binding
Footer zone	10–14 mm	Fixed	Page number and optional footer only
Header zone	20–28 mm	By template	Logo and page title hierarchy

4. Live Area and Content Zones

The live area is the region within which all essential learning content must remain. Each page is divided into zones so the Prompt Compiler™ can describe location and priority consistently.

Zone	Purpose	Constraint
Z01 — Brand Header	Top-left logo, series identity, page title	Never obstruct the page title or activity
Z02 — Learning Focus	Objective, target vocabulary or brief instruction	Maximum 15% of page area
Z03 — Main Illustration	Primary teacher-child learning scene	Usually 45–60% of page area
Z04 — Child Activity	Tracing, matching, colouring, speaking or response space	Must be physically usable
Z05 — Teacher Support	Teacher prompt or facilitation cue	Compact and visually distinct
Z06 — Parent Extension	Optional home connection	Small, non-dominant support box
Z07 — Footer	Page number, encouragement, rights text where required	Never carry core instruction

5. Visual-to-Text Ratio by Level

Level	Visual	Text	Production Interpretation
Nursery (3+)	70–80%	20–30%	Large objects, few instructions, strong white space
LKG (4+)	60–70%	30–40%	Richer scenes, short guided prompts
UKG (5+)	55–65%	35–45%	More structured activities and readiness cues
Grade 1+	45–60%	40–55%	Greater text density while retaining child-friendly hierarchy

6. Typography Production Rules

- Use highly legible rounded sans-serif typefaces approved by BCube brand governance.
- Never place body text directly over detailed artwork.
- Maintain strong hierarchy between page title, activity instruction, teacher guidance and parent extension.
- Avoid all-caps body instructions; reserve capitals for very short labels only.
- Minimum effective print size: 14 pt for Nursery child-facing text, 12 pt for LKG, 11 pt for UKG, unless tested and approved.
- Teacher and parent guidance may be smaller but must remain comfortably readable in print.
- Line spacing must prevent crowding; default 1.15–1.3 depending on typeface.

- Limit child instructions to one action per sentence whenever possible.

Element	Recommended Size	Weight	Maximum Length
Page title	24–34 pt	Bold	1–6 words
Learning objective	11–14 pt	Medium	18 words
Child instruction	14–20 pt	Semibold	12 words per instruction
Teacher prompt	10–12 pt	Regular	30 words
Parent extension	9.5–11 pt	Regular	35 words
Page number	9–11 pt	Regular	Numeric

7. Image and Illustration Resolution

- Raster artwork must be at least 300 pixels per inch at final placed size.
- Logos, simple icons and line art should use vector formats where possible.
- Do not upscale low-resolution images as a substitute for native-quality generation.
- Avoid compression artefacts, jagged outlines, muddy gradients and low-quality shadows.
- Every final illustration must preserve clean edges around characters, activity objects and text-safe zones.
- Generated images must not contain embedded unintended text, signatures, watermarks or unrelated logos.

8. Colour and Print Behaviour

- Design for reliable CMYK reproduction while keeping digital source files in a managed colour workflow.
- Use white or very light pastel page backgrounds for interior pages unless a special spread is approved.
- Avoid large areas of dense black or heavily saturated colour that increase ink load and reduce child readability.
- Maintain sufficient contrast between text and background.
- Do not depend on colour alone to communicate correct/incorrect, match groups or sequence.
- Test pale yellows, light blues and light greens for print visibility; subtle digital colours may disappear in print.
- Keep skin tones natural, diverse and consistent across pages.

9. White Space and Density Rules

White space is a learning aid. It separates instructions, reduces cognitive load and preserves a premium publishing appearance.

- Maintain visible breathing room around the page title, main scene and response area.

- Do not fill empty areas with decorative stars, balloons or icons unless they serve a learning purpose.
- Keep at least 6 mm between unrelated text boxes or visual components.
- Keep at least 8 mm around child writing, tracing or colouring spaces.
- A page must have one dominant learning focus; secondary decorative content may not compete for attention.
- Avoid more than three major content zones on Nursery activity pages unless the activity itself requires sequencing.

10. Cover Publishing Rules

- The cover must communicate subject, age level, series identity and emotional promise within three seconds.
- Official BCube logo placement follows current brand standards and remains clear of trim and character overlap.
- Age badge appears on the cover only unless a product brief states otherwise.
- Main title must remain readable at thumbnail size.
- Star mascot has a purposeful role related to the book theme rather than appearing as unrelated decoration.
- Do not overload the cover with more than one primary scene and three secondary learning cues.
- Spine and back cover are planned as part of the same production file.

11. Interior Brand Placement

- Official BCube logo appears in the approved header position on standard interior pages.
- Do not stretch, recolour, redraw or simplify the logo.
- Keep the logo visually subordinate to the page title and learning activity.
- Do not place the Nursery/LKG/UKG badge on standard interior activity pages.
- Page number appears consistently in the approved footer position.
- Series name, trademark and copyright wording must use approved phrasing.

12. Activity-Space Production Rules

- Tracing paths must be thick enough for the target age and leave room for imperfect pencil control.
- Colouring areas must be large, clearly enclosed and not broken by unnecessary fine detail.
- Matching activities must have unambiguous targets and sufficient line-drawing space.
- Cut-and-paste activities must show clear cut lines and safe adult-supervision cues where applicable.
- Writing areas must match the age level and expected response length.
- Do not let character hands, props, speech bubbles or decorative borders block the child response area.

13. Front Matter and Back Matter Rules

Page Type	Mandatory Purpose
Cover	Subject identity, level, logo, series, compelling visual
Copyright	Publisher, rights, edition, printing and legal details
Contents	Accurate page titles and page numbers
Welcome	Teacher/child orientation and book promise
Meet Star	Mascot role and tone
Review Pages	Mixed evidence, not decorative recap
Certificate	Child name, teacher signature, date, official branding
Back Cover	Series message, publisher identity, optional product summary

14. Accessibility and Inclusive Publishing

- Use clear layouts that can be understood even when a child cannot read every word.
- Avoid relying only on red/green colour distinctions.
- Keep instructions literal, concise and supported by visuals.
- Provide strong figure-ground separation between important objects and backgrounds.
- Represent diverse children without tokenism or stereotyping.
- Show adaptive participation naturally when relevant.
- Avoid tiny labels, dense text blocks and decorative scripts.

15. Production Rules for AI-Generated Artwork

- The prompt must explicitly reserve title, instruction and activity zones.
- AI-generated artwork must not render final instructional text unless the workflow supports controlled typography.
- All text should normally be added in the page-layout stage rather than inside the illustration.
- The prompt must prohibit watermarks, signatures, random labels and gibberish.
- The prompt must state the final page orientation, focal point, safe margins and activity-space requirements.
- Artwork is rejected if essential faces, hands, learning objects or activity areas are cropped.
- Artwork must be editable or composable enough to support final publishing layout.

16. Mandatory Negative Production Rules

- No watermarks, stock marks, signatures or unrelated logos.
- No dark full-page backgrounds for standard preschool interiors.
- No photorealistic children in the standard workbook illustration system.

- No distorted anatomy, extra fingers, missing limbs or cropped faces.
- No unsafe classroom setups, hazardous materials or unsupervised risky actions.
- No illegible text, gibberish, fake UI, random symbols or meaningless labels.
- No overcrowded layouts, tiny activity elements or decorative clutter.
- No important content outside the safe area.

17. Publishing Engine Input Schema

Field	Required	Example	Validation
product_level	Yes	Nursery 3+	Must match approved level list
page_type	Yes	Activity	Must match template category
page_number	Yes	12	Integer and sequence-valid
page_title	Yes	My Family	Exact approved title
orientation	Yes	A4 portrait	Must match product brief
visual_text_ratio	Yes	75:25	Within level range
activity_space	Conditional	Large colouring area	Required for response pages
logo_position	Yes	Top left	Approved brand position
footer_number	Yes	Bottom right	Consistent with template

18. Publishing Engine Output Block

The Publishing Engine™ emits a structured block that the Prompt Compiler™ inserts at the beginning of every final production prompt.

TECHNICAL AND PUBLISHING SPECIFICATION: Create an A4 portrait educational workbook page, 210 × 297 mm trim size, with 3 mm bleed, 300 DPI print resolution, clean white or approved pastel background, 10 mm safe margins, 12 mm inside binding margin, clear title zone, protected activity response area, official BCube logo in the approved header position, and page number in the approved footer position. Keep all essential text, faces, hands, learning objects and activity spaces inside the live area. Maintain the level-specific visual-to-text ratio, strong print contrast, uncluttered white space and commercial publishing quality.

19. Publishing Validation Checklist

☐ Correct trim size and orientation

☐ Bleed included where background reaches trim

☐ All essential content inside safe margins

☐ Binding margin protected

☐ Logo correctly positioned and unobstructed

☐ Title clearly readable

☐ Page number correctly positioned

☐ Visual-to-text ratio appropriate for level

☐ Child response area physically usable

☐ Typography large enough for print

☐ Illustration resolution suitable for final size

☐ Colour and contrast print-safe

☐ No embedded gibberish or unintended text

☐ No cropped faces, hands or learning objects

☐ No prohibited watermarks or unrelated logos

20. Gold Publishing Score

Category	Weight	Minimum Passing Score
Geometry and trim safety	20%	18/20
Readability and typography	15%	13/15
Activity usability	20%	18/20
Visual density and white space	15%	13/15
Brand placement	10%	9/10
Print and colour readiness	10%	9/10
Accessibility	10%	9/10

Passing rule: minimum total score 90/100, with no critical failure in trim safety, activity usability, logo integrity, accessibility or print resolution.

21. Compiler Rules

- The Publishing Engine block must appear before educational and illustration instructions.
- Page-specific geometry may override defaults only when an approved template ID is supplied.
- Conflicting downstream instructions are ignored when they violate safe margins, print resolution or brand placement.
- The Compiler must carry forward the exact page title and number from the page data schema.
- The final prompt must preserve a protected response area when the activity requires child output.

- The Publishing Engine version must be logged in the final prompt metadata.

22. Versioning and Change Control

- Major version: structural change to page geometry or production rules.
- Minor version: new template category, colour policy or layout capability.
- Patch version: clarification, correction or non-breaking validation update.
- Every release includes change summary, effective date, owner, affected products and migration guidance.
- Books already in production remain pinned to their approved engine version unless formally migrated.

23. Final Acceptance Criteria

Phase 1 is complete when the Publishing Engine™ can generate a consistent publishing specification for any approved BCube page type and every compiled prompt can be validated against the geometry, readability, activity, brand, accessibility and print-readiness rules in this document.

24. Phase 1 Deliverables

- Publishing Engine™ final specification
- A4 page geometry standard
- Live-area and zone definitions
- Typography production rules
- Visual-to-text ratios by level
- Cover and interior branding rules
- Activity-space standards
- AI artwork production constraints
- Publishing input schema
- Compiler-ready output block
- Validation checklist and Gold Publishing Score
- Version and change-control policy

Approval Statement

This document is adopted as the authoritative Phase 1 specification for BCube Prompt Engine™ v3.0. All future BCube Gold Production Prompts™ must inherit and comply with these publishing rules before illustration generation.

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Master_Part_2.docx`

Introduction

Part II begins the detailed rulebook for the Publishing Engine™. Each rule follows a normative structure with identifiers, requirements, rationale, validation, and compiler impact.

PE-GEO-001

Requirement: Every standard workbook page SHALL use the approved page geometry.

Rationale: Uniform production and printing.

Validation: Check page size, orientation and bleed.

Compiler Impact: Publishing block generated.

PE-GEO-002

Requirement: Essential content MUST remain inside the safe area.

Rationale: Prevents trim loss.

Validation: Measure all critical elements against safe margin.

Compiler Impact: Safe-area flag.

PE-LAY-001

Requirement: Pages SHALL reserve protected zones for title, activity and footer.

Rationale: Supports consistent layouts.

Validation: Template compliance review.

Compiler Impact: Layout metadata.

PE-TYP-001

Requirement: Child-facing typography MUST meet minimum readability standards.

Rationale: Supports early readers.

Validation: Font size and contrast validation.

Compiler Impact: Typography block.

PE-CLR-001

Requirement: Interior pages SHOULD use light backgrounds with accessible contrast.

Rationale: Improves readability.

Validation: Contrast review.

Compiler Impact: Colour profile.

PE-IMG-001

Requirement: Illustrations MUST be suitable for 300 DPI print reproduction.

Rationale: Commercial print quality.

Validation: Resolution validation.

Compiler Impact: Image compliance.

PE-BRD-001

Requirement: Official BCube branding SHALL follow approved placement rules.

Rationale: Brand consistency.

Validation: Header inspection.

Compiler Impact: Brand metadata.

PE-ACT-001

Requirement: Activity spaces MUST remain unobstructed.

Rationale: Supports child interaction.

Validation: Activity-zone validation.

Compiler Impact: Activity metadata.

PE-EXP-001

Requirement: Final output SHALL support PDF/X export workflow.

Rationale: Reliable production.

Validation: Export profile verification.

Compiler Impact: Release metadata.

PE-VAL-001

Requirement: Publishing Engine SHALL fail compilation on critical geometry defects.

Rationale: Protects production quality.

Validation: Critical error detection.

Compiler Impact: Compilation stop.

Publishing Engine Prompt Fragment

TECHNICAL SPECIFICATION: • Approved page geometry • Safe margins • Bleed • Title zone • Activity zone • Footer zone • 300 DPI • Print-ready composition • Accessible typography • Protected child response areas

Expansion Plan

Subsequent sections will expand each rule into detailed parameter tables, edge cases, exceptions, examples, validation algorithms, reusable prompt fragments and machine-readable compiler definitions.

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Master_Part_3.docx`

PE-GEO-001 Detailed Specification

This section expands the mandatory page geometry rule into an operational standard suitable for human authors, designers, AI systems and automated validation.

Objective

Ensure every page is physically manufacturable and visually consistent.

Scope

Applies to all workbook interiors, assessments, certificates and teacher resources unless an approved exception exists.

Normative Requirement

Every production prompt SHALL inherit the approved geometry profile before any design or illustration instructions are applied.

Inputs

Book level, page type, trim profile, template ID, binding type.

Outputs

Normalized publishing metadata including page size, bleed, safe area, live area and protected zones.

Dependencies

Publishing Engine is executed before Design Engine and cannot be bypassed.

Failure Conditions

Missing geometry profile, incorrect orientation, protected content outside safe area, inconsistent template assignment.

Compiler Behaviour

Compilation stops on critical geometry failures and reports the affected rule identifier.

Operational Parameters

Parameter	Default	Allowed Range	Validation
Trim Size	210×297 mm	Product profile	Exact match required
Orientation	Portrait	Portrait unless approved	Reject mismatch
Bleed	3 mm	3–5 mm	Warn if >5 mm
Safe Margin	10 mm	8–15 mm	Reject below minimum
Binding Margin	12 mm	12–18 mm	Adjust by binding
Live Area	Computed	Derived	Auto-generated
Header Zone	Reserved	Template driven	Required
Footer Zone	Reserved	Template driven	Required

Validation Algorithm

Step 1: Load template profile. Step 2: Resolve trim size. Step 3: Calculate bleed and safe area. Step 4: Validate all protected content. Step 5: Validate header, footer and activity zones. Step 6: Emit publishing metadata. Step 7: Pass to Design Engine™.

Good Practices

- Reserve generous whitespace around child response areas.
- Keep essential faces, hands and learning objects within the live area.
- Allow for binding creep on multi-page books.
- Validate every exported page before illustration approval.

Anti-Patterns

- Placing page numbers in variable locations.
- Allowing artwork to hide activity spaces.
- Using inconsistent margins between templates.
- Cropping educational objects at trim edges.

Cross References

Related rules: PE-GEO-002, PE-LAY-001, DE-GRID-001, VG-FOC-001, QA-PUB-001.

Expansion Note

This pattern will be repeated for every rule in the BCube Prompt Engine™. Each rule will eventually include detailed examples, exception cases, prompt fragments, automated validation logic and implementation notes.

Phase 2 - Design Engine

Phase 2 - Design Engine

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Purpose

Design Engine™ transforms the publishing specification into a consistent page layout system. It defines visual hierarchy, reusable layout components, spacing, typography application, iconography, component placement, and template behavior so every BCube page follows one coherent design language.

1. Design Principles

Objective: Standardize this design layer across every BCube publication.

- Child-first clarity
- One dominant learning focus
- Consistency before decoration
- Purposeful white space
- Accessible layouts
- Premium publishing appearance

Compiler Output

Produces a normalized design specification describing page grid, component positions, spacing rules, typography hierarchy and template identifiers for downstream engines.

Acceptance Criteria

All pages exhibit consistent layout, visual hierarchy, spacing, component placement and design language regardless of book or subject.

2. Layout Grid System

Objective: Standardize this design layer across every BCube publication.

- 12-column responsive print grid
- Header, content and footer regions
- Consistent gutters
- Template locking
- Alignment rules
- Baseline spacing

Compiler Output

Produces a normalized design specification describing page grid, component positions, spacing rules, typography hierarchy and template identifiers for downstream engines.

Acceptance Criteria

All pages exhibit consistent layout, visual hierarchy, spacing, component placement and design language regardless of book or subject.

3. Visual Hierarchy

Objective: Standardize this design layer across every BCube publication.

- Title > Illustration > Activity > Teacher prompt > Parent prompt > Footer
- Single primary focal area
- Limit competing visual elements

Compiler Output

Produces a normalized design specification describing page grid, component positions, spacing rules, typography hierarchy and template identifiers for downstream engines.

Acceptance Criteria

All pages exhibit consistent layout, visual hierarchy, spacing, component placement and design language regardless of book or subject.

4. Component Library

Objective: Standardize this design layer across every BCube publication.

- Page title
- Learning objective
- Instruction box
- Teacher tip
- Parent extension
- Activity panel
- Reward badge
- Footer
- Icons

Compiler Output

Produces a normalized design specification describing page grid, component positions, spacing rules, typography hierarchy and template identifiers for downstream engines.

Acceptance Criteria

All pages exhibit consistent layout, visual hierarchy, spacing, component placement and design language regardless of book or subject.

5. Typography Application

Objective: Standardize this design layer across every BCube publication.

- Heading scale
- Instruction scale
- Consistent line spacing

- Readable contrast
- Maximum line lengths

Compiler Output

Produces a normalized design specification describing page grid, component positions, spacing rules, typography hierarchy and template identifiers for downstream engines.

Acceptance Criteria

All pages exhibit consistent layout, visual hierarchy, spacing, component placement and design language regardless of book or subject.

6. Colour & UI Tokens

Objective: Standardize this design layer across every BCube publication.

- Primary brand colours
- Secondary learning colours
- Semantic colours
- Neutral palette
- Accessibility contrast

Compiler Output

Produces a normalized design specification describing page grid, component positions, spacing rules, typography hierarchy and template identifiers for downstream engines.

Acceptance Criteria

All pages exhibit consistent layout, visual hierarchy, spacing, component placement and design language regardless of book or subject.

7. Iconography Standards

Objective: Standardize this design layer across every BCube publication.

- Simple rounded icons
- Consistent stroke width
- Meaningful symbols
- No decorative-only icons

Compiler Output

Produces a normalized design specification describing page grid, component positions, spacing rules, typography hierarchy and template identifiers for downstream engines.

Acceptance Criteria

All pages exhibit consistent layout, visual hierarchy, spacing, component placement and design language regardless of book or subject.

8. Template System

Objective: Standardize this design layer across every BCube publication.

- Cover template
- Activity template
- Assessment template
- Certificate template
- Review template
- Story template

Compiler Output

Produces a normalized design specification describing page grid, component positions, spacing rules, typography hierarchy and template identifiers for downstream engines.

Acceptance Criteria

All pages exhibit consistent layout, visual hierarchy, spacing, component placement and design language regardless of book or subject.

9. Responsive Composition Rules

Objective: Standardize this design layer across every BCube publication.

- Balance illustration and activity
- Reserve response space
- Protect safe zones
- Avoid overlap

Compiler Output

Produces a normalized design specification describing page grid, component positions, spacing rules, typography hierarchy and template identifiers for downstream engines.

Acceptance Criteria

All pages exhibit consistent layout, visual hierarchy, spacing, component placement and design language regardless of book or subject.

10. Design Validation

Objective: Standardize this design layer across every BCube publication.

- Alignment
- Spacing
- Readability
- Brand consistency
- Template compliance
- Print readiness

Compiler Output

Produces a normalized design specification describing page grid, component positions, spacing rules, typography hierarchy and template identifiers for downstream engines.

Acceptance Criteria

All pages exhibit consistent layout, visual hierarchy, spacing, component placement and design language regardless of book or subject.

Standard Design Tokens

Token	Purpose	Example
SP-8	Base spacing	8 mm
GRID-12	Layout grid	12-column
TITLE-H1	Primary title	Page title
ACT-ZONE	Activity area	Tracing/Matching
ILLUS-PRIMARY	Main illustration	45–60% page
FOOTER-STD	Footer	Page number

Compiler Sequence

Publishing Engine™ → Design Engine™ → Visual Grammar Engine™ → Educational Engine™ → Teaching Engine™ → Illustration Engine™ → QA Engine™ → Prompt Compiler™

Phase 2 Deliverables

- BCube Design Language™ specification
- Grid system
- Component library
- Template library
- Design tokens
- Visual hierarchy rules
- Layout validation checklist
- Compiler-ready design block

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Master_Part_4.docx`

Purpose

This part expands the Design Engine™ into a normative engineering specification. It defines reusable layout rules, design tokens, grid behavior, spacing, typography application, component placement and validation requirements.

DE-GRID-001

Requirement: Every page SHALL be based on an approved BCube layout grid.

Rationale: Creates consistent, reusable page structures across every BCube publication.

Inputs: Publishing metadata, template ID, design tokens, page type.

Outputs: Resolved layout, component positions, spacing and typography metadata.

Validation: Check grid alignment, spacing consistency, template compliance and hierarchy.

Compiler Impact: Provides normalized design instructions for downstream engines.

DE-GRID-002

Requirement: Components MUST snap to the defined grid and spacing tokens.

Rationale: Creates consistent, reusable page structures across every BCube publication.

Inputs: Publishing metadata, template ID, design tokens, page type.

Outputs: Resolved layout, component positions, spacing and typography metadata.

Validation: Check grid alignment, spacing consistency, template compliance and hierarchy.

Compiler Impact: Provides normalized design instructions for downstream engines.

DE-HIER-001

Requirement: A page SHALL have one dominant visual hierarchy.

Rationale: Creates consistent, reusable page structures across every BCube publication.

Inputs: Publishing metadata, template ID, design tokens, page type.

Outputs: Resolved layout, component positions, spacing and typography metadata.

Validation: Check grid alignment, spacing consistency, template compliance and hierarchy.

Compiler Impact: Provides normalized design instructions for downstream engines.

DE-COMP-001

Requirement: Every page SHALL use only approved UI/page components.

Rationale: Creates consistent, reusable page structures across every BCube publication.

Inputs: Publishing metadata, template ID, design tokens, page type.

Outputs: Resolved layout, component positions, spacing and typography metadata.

Validation: Check grid alignment, spacing consistency, template compliance and hierarchy.

Compiler Impact: Provides normalized design instructions for downstream engines.

DE-SPC-001

Requirement: Whitespace MUST preserve readability and activity usability.

Rationale: Creates consistent, reusable page structures across every BCube publication.

Inputs: Publishing metadata, template ID, design tokens, page type.

Outputs: Resolved layout, component positions, spacing and typography metadata.

Validation: Check grid alignment, spacing consistency, template compliance and hierarchy.

Compiler Impact: Provides normalized design instructions for downstream engines.

DE-TOK-001

Requirement: Only approved BCube design tokens MAY be used.

Rationale: Creates consistent, reusable page structures across every BCube publication.

Inputs: Publishing metadata, template ID, design tokens, page type.

Outputs: Resolved layout, component positions, spacing and typography metadata.

Validation: Check grid alignment, spacing consistency, template compliance and hierarchy.

Compiler Impact: Provides normalized design instructions for downstream engines.

DE-TEMP-001

Requirement: Each page SHALL inherit an approved template profile.

Rationale: Creates consistent, reusable page structures across every BCube publication.

Inputs: Publishing metadata, template ID, design tokens, page type.

Outputs: Resolved layout, component positions, spacing and typography metadata.

Validation: Check grid alignment, spacing consistency, template compliance and hierarchy.

Compiler Impact: Provides normalized design instructions for downstream engines.

DE-TYPE-001

Requirement: Typography SHALL follow BCube hierarchy and readability rules.

Rationale: Creates consistent, reusable page structures across every BCube publication.

Inputs: Publishing metadata, template ID, design tokens, page type.

Outputs: Resolved layout, component positions, spacing and typography metadata.

Validation: Check grid alignment, spacing consistency, template compliance and hierarchy.

Compiler Impact: Provides normalized design instructions for downstream engines.

DE-RESP-001

Requirement: Layouts MUST preserve protected activity zones.

Rationale: Creates consistent, reusable page structures across every BCube publication.

Inputs: Publishing metadata, template ID, design tokens, page type.

Outputs: Resolved layout, component positions, spacing and typography metadata.

Validation: Check grid alignment, spacing consistency, template compliance and hierarchy.

Compiler Impact: Provides normalized design instructions for downstream engines.

DE-VAL-001

Requirement: Design validation MUST complete before Visual Grammar compilation.

Rationale: Creates consistent, reusable page structures across every BCube publication.

Inputs: Publishing metadata, template ID, design tokens, page type.

Outputs: Resolved layout, component positions, spacing and typography metadata.

Validation: Check grid alignment, spacing consistency, template compliance and hierarchy.

Compiler Impact: Provides normalized design instructions for downstream engines.

Design Token Catalogue

Token	Purpose	Default
GRID-12	Layout grid	12-column
SPACE-08	Base spacing	8 mm
SPACE-16	Section spacing	16 mm
TITLE-H1	Primary heading	Page title
BODY-01	Instruction text	Child readable
ACT-AREA	Activity region	Protected
SAFE-ZONE	Live content	Mandatory

Design Anti-Patterns

- Components floating off-grid
- Inconsistent spacing between pages
- Multiple competing headlines
- Decorative elements inside activity zones
- Typography overlapping illustrations
- Unapproved colours or icons

Cross References

Related specifications: PE-LAY-001, VG-FOC-001, EE-OBJ-001, QA-DES-001.

Next Expansion

Subsequent revisions will expand each Design Engine™ rule with diagrams, parameter tables, examples, edge cases, validation algorithms and reusable compiler fragments.

Phase 3 - Visual Grammar Engine

Phase 3 - Visual Grammar Engine

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Purpose

Visual Grammar Engine™ defines how children visually read a BCube page before they can read text. It standardizes composition, eye flow, focal hierarchy, visual rhythm, cognitive load, whitespace, scene balance and instructional emphasis so illustrations consistently support learning.

1. Visual Philosophy

Objective: Define mandatory visual composition standards for all BCube illustrations.

- Learning before decoration
- One clear learning focus
- Emotion supports comprehension
- Every visual element has instructional purpose
- Consistent visual language

Compiler Output

Outputs visual composition metadata including focal hierarchy, eye-flow path, scene balance, whitespace allocation and attention priorities.

Acceptance Criteria

Children can immediately identify the learning focus, follow the intended visual sequence and complete the activity without visual confusion.

2. Eye Flow Architecture

Objective: Define mandatory visual composition standards for all BCube illustrations.

- Top-to-bottom guidance
- Left-to-right sequencing where appropriate
- Natural gaze path
- Avoid conflicting attention anchors
- End gaze at child activity area

Compiler Output

Outputs visual composition metadata including focal hierarchy, eye-flow path, scene balance, whitespace allocation and attention priorities.

Acceptance Criteria

Children can immediately identify the learning focus, follow the intended visual sequence and complete the activity without visual confusion.

3. Focal Point Rules

Objective: Define mandatory visual composition standards for all BCube illustrations.

- One primary focal point
- Maximum two secondary focal points
- Teacher and Star reinforce—not compete with—the objective
- Learning object always remains prominent

Compiler Output

Outputs visual composition metadata including focal hierarchy, eye-flow path, scene balance, whitespace allocation and attention priorities.

Acceptance Criteria

Children can immediately identify the learning focus, follow the intended visual sequence and complete the activity without visual confusion.

4. Composition Framework

Objective: Define mandatory visual composition standards for all BCube illustrations.

- Rule of thirds
- Balanced negative space
- Foreground, middle ground and background
- Clear depth without clutter
- Safe cropping rules

Compiler Output

Outputs visual composition metadata including focal hierarchy, eye-flow path, scene balance, whitespace allocation and attention priorities.

Acceptance Criteria

Children can immediately identify the learning focus, follow the intended visual sequence and complete the activity without visual confusion.

5. Cognitive Load Standards

Objective: Define mandatory visual composition standards for all BCube illustrations.

- Limit simultaneous concepts
- Reduce decorative noise
- Chunk information
- Simple visual grouping
- Age-appropriate complexity

Compiler Output

Outputs visual composition metadata including focal hierarchy, eye-flow path, scene balance, whitespace allocation and attention priorities.

Acceptance Criteria

Children can immediately identify the learning focus, follow the intended visual sequence and complete the activity without visual confusion.

6. White Space Strategy

Objective: Define mandatory visual composition standards for all BCube illustrations.

- Breathing room around activities
- Protected instruction zones
- Clear separation between unrelated tasks
- Whitespace as a learning aid

Compiler Output

Outputs visual composition metadata including focal hierarchy, eye-flow path, scene balance, whitespace allocation and attention priorities.

Acceptance Criteria

Children can immediately identify the learning focus, follow the intended visual sequence and complete the activity without visual confusion.

7. Scene Storytelling

Objective: Define mandatory visual composition standards for all BCube illustrations.

- Beginning → interaction → learning moment → positive outcome
- Meaningful expressions
- Natural classroom interactions

Compiler Output

Outputs visual composition metadata including focal hierarchy, eye-flow path, scene balance, whitespace allocation and attention priorities.

Acceptance Criteria

Children can immediately identify the learning focus, follow the intended visual sequence and complete the activity without visual confusion.

8. Visual Consistency

Objective: Define mandatory visual composition standards for all BCube illustrations.

- Consistent scale
- Lighting
- Perspective
- Colour harmony
- Object proportions
- Recurring environments

Compiler Output

Outputs visual composition metadata including focal hierarchy, eye-flow path, scene balance, whitespace allocation and attention priorities.

Acceptance Criteria

Children can immediately identify the learning focus, follow the intended visual sequence and complete the activity without visual confusion.

9. Attention Mapping

Objective: Define mandatory visual composition standards for all BCube illustrations.

- Title
- Main illustration
- Learning object
- Activity
- Teacher cue
- Parent cue
- Footer

Compiler Output

Outputs visual composition metadata including focal hierarchy, eye-flow path, scene balance, whitespace allocation and attention priorities.

Acceptance Criteria

Children can immediately identify the learning focus, follow the intended visual sequence and complete the activity without visual confusion.

10. Validation Engine

Objective: Define mandatory visual composition standards for all BCube illustrations.

- Focal clarity
- Eye flow
- Readability
- Instruction visibility
- Scene balance
- Educational effectiveness

Compiler Output

Outputs visual composition metadata including focal hierarchy, eye-flow path, scene balance, whitespace allocation and attention priorities.

Acceptance Criteria

Children can immediately identify the learning focus, follow the intended visual sequence and complete the activity without visual confusion.

Visual Hierarchy Matrix

Priority	Element	Purpose
1	Page Title	Orient the learner
2	Primary Illustration	Introduce concept
3	Learning Object	Teach target concept
4	Activity Area	Child participation
5	Teacher Cue	Support facilitation
6	Parent Cue	Home extension
7	Footer	Navigation

Negative Visual Rules

- No competing focal points
- No overcrowded scenes
- No tiny instructional objects
- No decorative clutter
- No hidden learning object
- No excessive background detail
- No obstructed activity space
- No unrelated visual elements

Phase 3 Deliverables

- BCube Visual Grammar™ Specification
- Eye Flow Rules
- Composition Framework
- Attention Mapping Model
- Cognitive Load Standards
- Whitespace Standards
- Visual Validation Checklist
- Compiler-ready Visual Grammar Block

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Master_Part_5.docx`

Purpose

This section expands the Visual Grammar Engine™ into a rule-based specification governing composition, eye flow, focal hierarchy, visual rhythm, cognitive load and instructional clarity.

VG-FOC-001

Requirement: Every page SHALL contain exactly one dominant educational focal point.

Rationale: Improve comprehension and reduce visual confusion.

Inputs: Design layout, illustration metadata, page objective.

Outputs: Focal hierarchy, eye-flow map, composition metadata.

Validation: Evaluate attention path, clutter, balance and instructional clarity.

Compiler Impact: Generates normalized visual grammar block for prompt compilation.

VG-FOC-002

Requirement: Secondary focal points MUST reinforce, not compete with, the primary objective.

Rationale: Improve comprehension and reduce visual confusion.

Inputs: Design layout, illustration metadata, page objective.

Outputs: Focal hierarchy, eye-flow map, composition metadata.

Validation: Evaluate attention path, clutter, balance and instructional clarity.

Compiler Impact: Generates normalized visual grammar block for prompt compilation.

VG-EYE-001

Requirement: Visual eye flow SHALL guide the learner naturally from title to activity.

Rationale: Improve comprehension and reduce visual confusion.

Inputs: Design layout, illustration metadata, page objective.

Outputs: Focal hierarchy, eye-flow map, composition metadata.

Validation: Evaluate attention path, clutter, balance and instructional clarity.

Compiler Impact: Generates normalized visual grammar block for prompt compilation.

VG-BAL-001

Requirement: Visual weight SHALL remain balanced across the page.

Rationale: Improve comprehension and reduce visual confusion.

Inputs: Design layout, illustration metadata, page objective.

Outputs: Focal hierarchy, eye-flow map, composition metadata.

Validation: Evaluate attention path, clutter, balance and instructional clarity.

Compiler Impact: Generates normalized visual grammar block for prompt compilation.

VG-RHY-001

Requirement: Spacing and repetition SHOULD create predictable visual rhythm.

Rationale: Improve comprehension and reduce visual confusion.

Inputs: Design layout, illustration metadata, page objective.

Outputs: Focal hierarchy, eye-flow map, composition metadata.

Validation: Evaluate attention path, clutter, balance and instructional clarity.

Compiler Impact: Generates normalized visual grammar block for prompt compilation.

VG-COG-001

Requirement: Illustrations MUST minimize unnecessary cognitive load.

Rationale: Improve comprehension and reduce visual confusion.

Inputs: Design layout, illustration metadata, page objective.

Outputs: Focal hierarchy, eye-flow map, composition metadata.

Validation: Evaluate attention path, clutter, balance and instructional clarity.

Compiler Impact: Generates normalized visual grammar block for prompt compilation.

VG-WHT-001

Requirement: Whitespace SHALL be treated as an instructional design element.

Rationale: Improve comprehension and reduce visual confusion.

Inputs: Design layout, illustration metadata, page objective.

Outputs: Focal hierarchy, eye-flow map, composition metadata.

Validation: Evaluate attention path, clutter, balance and instructional clarity.

Compiler Impact: Generates normalized visual grammar block for prompt compilation.

VG-SCN-001

Requirement: Scenes MUST communicate a single learning narrative.

Rationale: Improve comprehension and reduce visual confusion.

Inputs: Design layout, illustration metadata, page objective.

Outputs: Focal hierarchy, eye-flow map, composition metadata.

Validation: Evaluate attention path, clutter, balance and instructional clarity.

Compiler Impact: Generates normalized visual grammar block for prompt compilation.

VG-ACC-001

Requirement: Visual grammar SHALL support accessibility and inclusion.

Rationale: Improve comprehension and reduce visual confusion.

Inputs: Design layout, illustration metadata, page objective.

Outputs: Focal hierarchy, eye-flow map, composition metadata.

Validation: Evaluate attention path, clutter, balance and instructional clarity.

Compiler Impact: Generates normalized visual grammar block for prompt compilation.

VG-VAL-001

Requirement: Visual Grammar validation MUST complete before Educational Engine execution.

Rationale: Improve comprehension and reduce visual confusion.

Inputs: Design layout, illustration metadata, page objective.

Outputs: Focal hierarchy, eye-flow map, composition metadata.

Validation: Evaluate attention path, clutter, balance and instructional clarity.

Compiler Impact: Generates normalized visual grammar block for prompt compilation.

Visual Hierarchy Matrix

Priority	Element	Rule
1	Page Title	Immediate orientation
2	Primary Learning Illustration	Dominant focal point
3	Target Learning Object	Supports objective
4	Child Activity	Action destination
5	Teacher Cue	Secondary guidance
6	Parent Cue	Optional extension

Visual Anti-Patterns

- Multiple competing focal points
- Busy backgrounds masking learning objects
- Decorative elements overpowering instruction
- Inconsistent visual scale
- Hidden activity areas
- Visual noise without educational purpose

Cross References

Related rules: DE-HIER-001, IE-SCN-001, QA-VIS-001, PC-ASM-001.

Expansion Roadmap

Future revisions will expand each Visual Grammar rule with measurable parameters, annotated examples, edge cases, validation algorithms, prompt fragments and compiler schemas.

Phase 4 - Educational Engine

Phase 4 - Educational Engine

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Purpose

The Educational Engine™ is responsible for embedding learning science into every BCube prompt. It ensures every illustration, activity and interaction has a measurable educational purpose aligned with BCube EduOS™, curriculum progression and age-appropriate developmental outcomes.

1. Educational Philosophy

Objective: Define mandatory educational rules applied to every compiled BCube production prompt.

- Child-first learning
- Learning through play
- Competency-based progression
- Whole-child development
- Future-ready skills
- Joyful learning

Compiler Output

Produces structured educational metadata including learning objective, competency map, developmental level, assessment evidence and expected child outcome.

Acceptance Criteria

Every generated page demonstrates a clear educational purpose, measurable learning outcome and alignment with BCube curriculum standards.

2. Learning Objective Framework

Objective: Define mandatory educational rules applied to every compiled BCube production prompt.

- One primary objective per page
- Optional supporting objective
- Observable learning outcome
- Clear success criteria
- Curriculum alignment

Compiler Output

Produces structured educational metadata including learning objective, competency map, developmental level, assessment evidence and expected child outcome.

Acceptance Criteria

Every generated page demonstrates a clear educational purpose, measurable learning outcome and alignment with BCube curriculum standards.

3. Competency Mapping

Objective: Define mandatory educational rules applied to every compiled BCube production prompt.

- Curiosity
- Communication
- Creativity
- Critical Thinking
- Collaboration
- Confidence
- Character

Compiler Output

Produces structured educational metadata including learning objective, competency map, developmental level, assessment evidence and expected child outcome.

Acceptance Criteria

Every generated page demonstrates a clear educational purpose, measurable learning outcome and alignment with BCube curriculum standards.

4. Developmental Alignment

Objective: Define mandatory educational rules applied to every compiled BCube production prompt.

- Age appropriateness
- Nursery progression
- LKG progression
- UKG progression
- Grade 1 readiness

Compiler Output

Produces structured educational metadata including learning objective, competency map, developmental level, assessment evidence and expected child outcome.

Acceptance Criteria

Every generated page demonstrates a clear educational purpose, measurable learning outcome and alignment with BCube curriculum standards.

5. Activity Design Standards

Objective: Define mandatory educational rules applied to every compiled BCube production prompt.

- Hands-on engagement
- Talk before write
- Explore before explain
- One core action
- Meaningful repetition

Compiler Output

Produces structured educational metadata including learning objective, competency map, developmental level, assessment evidence and expected child outcome.

Acceptance Criteria

Every generated page demonstrates a clear educational purpose, measurable learning outcome and alignment with BCube curriculum standards.

6. Assessment Evidence

Objective: Define mandatory educational rules applied to every compiled BCube production prompt.

- Observable behaviour
- Teacher evidence
- Child artefact
- Conversation evidence
- Completion criteria

Compiler Output

Produces structured educational metadata including learning objective, competency map, developmental level, assessment evidence and expected child outcome.

Acceptance Criteria

Every generated page demonstrates a clear educational purpose, measurable learning outcome and alignment with BCube curriculum standards.

7. Differentiation

Objective: Define mandatory educational rules applied to every compiled BCube production prompt.

- Core pathway
- Support pathway
- Extension pathway
- Inclusive participation
- Flexible pacing

Compiler Output

Produces structured educational metadata including learning objective, competency map, developmental level, assessment evidence and expected child outcome.

Acceptance Criteria

Every generated page demonstrates a clear educational purpose, measurable learning outcome and alignment with BCube curriculum standards.

8. Learning Transfer

Objective: Define mandatory educational rules applied to every compiled BCube production prompt.

- Home connection
- Real-world example
- Cross-curricular links
- Daily life application

Compiler Output

Produces structured educational metadata including learning objective, competency map, developmental level, assessment evidence and expected child outcome.

Acceptance Criteria

Every generated page demonstrates a clear educational purpose, measurable learning outcome and alignment with BCube curriculum standards.

9. Metadata Schema

Objective: Define mandatory educational rules applied to every compiled BCube production prompt.

- Book
- Unit
- Lesson
- Objective
- Competency
- Vocabulary
- Materials
- Assessment
- Future skill

Compiler Output

Produces structured educational metadata including learning objective, competency map, developmental level, assessment evidence and expected child outcome.

Acceptance Criteria

Every generated page demonstrates a clear educational purpose, measurable learning outcome and alignment with BCube curriculum standards.

10. Educational Validation

Objective: Define mandatory educational rules applied to every compiled BCube production prompt.

- Objective visible
- Competency represented
- Age appropriate
- Assessment possible
- Learning measurable
- EduOS compliance

Compiler Output

Produces structured educational metadata including learning objective, competency map, developmental level, assessment evidence and expected child outcome.

Acceptance Criteria

Every generated page demonstrates a clear educational purpose, measurable learning outcome and alignment with BCube curriculum standards.

Educational Metadata Schema

Field	Description	Required
Learning Objective	Primary instructional goal	Yes
Competency	BCube Learning DNA mapping	Yes
Age Level	Target learner	Yes
Vocabulary	Key language	Optional
Future Skill	Transfer skill	Yes
Assessment Evidence	Observable success	Yes
Differentiation	Support/extension	Optional
Home Extension	Parent connection	Optional

Negative Educational Rules

- No page without a measurable learning objective
- No multiple unrelated concepts on one page
- No activity without observable evidence
- No abstract language beyond developmental level
- No decorative content replacing learning intent
- No assessment that exceeds age capability

Phase 4 Deliverables

- Educational Engine™ specification
- Learning Objective Framework

- Competency Mapping Guide
- Developmental Alignment Matrix
- Assessment Metadata Schema
- Differentiation Rules
- Educational Validation Checklist
- Compiler-ready Educational Block

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Master_Part_6.docx`

Purpose

This section formalizes the Educational Engine™ as a standards-based rule system governing curriculum alignment, learning objectives, competencies, developmental progression, activity design, assessment evidence and learning transfer.

EE-OBJ-001

Requirement: Every page SHALL define one measurable primary learning objective.

Rationale: Ensure educational consistency and measurable learning outcomes.

Inputs: Curriculum, age level, competencies, page metadata.

Outputs: Structured educational metadata for downstream engines.

Validation: Verify objective clarity, competency alignment and assessment readiness.

Compiler Impact: Generates Educational Engine block for the final BCube Gold Production Prompt™.

EE-OBJ-002

Requirement: Supporting objectives MUST reinforce the primary objective.

Rationale: Ensure educational consistency and measurable learning outcomes.

Inputs: Curriculum, age level, competencies, page metadata.

Outputs: Structured educational metadata for downstream engines.

Validation: Verify objective clarity, competency alignment and assessment readiness.

Compiler Impact: Generates Educational Engine block for the final BCube Gold Production Prompt™.

EE-CMP-001

Requirement: Each activity SHALL map to one or more BCube competencies.

Rationale: Ensure educational consistency and measurable learning outcomes.

Inputs: Curriculum, age level, competencies, page metadata.

Outputs: Structured educational metadata for downstream engines.

Validation: Verify objective clarity, competency alignment and assessment readiness.

Compiler Impact: Generates Educational Engine block for the final BCube Gold Production Prompt™.

EE-DEV-001

Requirement: Learning experiences MUST align with the target developmental stage.

Rationale: Ensure educational consistency and measurable learning outcomes.

Inputs: Curriculum, age level, competencies, page metadata.

Outputs: Structured educational metadata for downstream engines.

Validation: Verify objective clarity, competency alignment and assessment readiness.

Compiler Impact: Generates Educational Engine block for the final BCube Gold Production Prompt™.

EE-ACT-001

Requirement: Activities SHALL require meaningful child participation.

Rationale: Ensure educational consistency and measurable learning outcomes.

Inputs: Curriculum, age level, competencies, page metadata.

Outputs: Structured educational metadata for downstream engines.

Validation: Verify objective clarity, competency alignment and assessment readiness.

Compiler Impact: Generates Educational Engine block for the final BCube Gold Production Prompt™.

EE-ASM-001

Requirement: Every activity MUST include observable assessment evidence.

Rationale: Ensure educational consistency and measurable learning outcomes.

Inputs: Curriculum, age level, competencies, page metadata.

Outputs: Structured educational metadata for downstream engines.

Validation: Verify objective clarity, competency alignment and assessment readiness.

Compiler Impact: Generates Educational Engine block for the final BCube Gold Production Prompt™.

EE-DIF-001

Requirement: Support and extension pathways SHOULD be available.

Rationale: Ensure educational consistency and measurable learning outcomes.

Inputs: Curriculum, age level, competencies, page metadata.

Outputs: Structured educational metadata for downstream engines.

Validation: Verify objective clarity, competency alignment and assessment readiness.

Compiler Impact: Generates Educational Engine block for the final BCube Gold Production Prompt™.

EE-TRN-001

Requirement: Learning SHOULD transfer to real-life experiences.

Rationale: Ensure educational consistency and measurable learning outcomes.

Inputs: Curriculum, age level, competencies, page metadata.

Outputs: Structured educational metadata for downstream engines.

Validation: Verify objective clarity, competency alignment and assessment readiness.

Compiler Impact: Generates Educational Engine block for the final BCube Gold Production Prompt™.

EE-MTD-001

Requirement: Educational metadata SHALL be complete before compilation.

Rationale: Ensure educational consistency and measurable learning outcomes.

Inputs: Curriculum, age level, competencies, page metadata.

Outputs: Structured educational metadata for downstream engines.

Validation: Verify objective clarity, competency alignment and assessment readiness.

Compiler Impact: Generates Educational Engine block for the final BCube Gold Production Prompt™.

EE-VAL-001

Requirement: Educational validation MUST pass before illustration approval.

Rationale: Ensure educational consistency and measurable learning outcomes.

Inputs: Curriculum, age level, competencies, page metadata.

Outputs: Structured educational metadata for downstream engines.

Validation: Verify objective clarity, competency alignment and assessment readiness.

Compiler Impact: Generates Educational Engine block for the final BCube Gold Production Prompt™.

Educational Metadata Matrix

Field	Purpose	Required
Primary Learning Objective	Defines expected learning	Yes
Competency	Maps to BCube Learning DNA	Yes
Developmental Stage	Age alignment	Yes
Assessment Evidence	Observable success	Yes
Future Skill	Transfer capability	Yes
Vocabulary	Language support	Optional

Field	Purpose	Required
Differentiation	Support and extension	Optional

Educational Anti-Patterns

- Multiple unrelated objectives on one page
- Activities without measurable outcomes
- Developmentally inappropriate expectations
- Assessment disconnected from objectives
- Vocabulary exceeding learner readiness
- Learning without real-world relevance

Cross References

Related rules: VG-SCN-001, TE-INT-001, PP-HME-001, QA-EDU-001.

Expansion Roadmap

Future revisions will expand each Educational Engine rule into detailed parameter specifications, developmental matrices, worked examples, assessment rubrics, validation algorithms, reusable prompt fragments and compiler schemas.

Phase 5 - Teaching Engine

Phase 5 - Teaching Engine

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Purpose

The Teaching Engine™ embeds effective classroom pedagogy into every BCube production prompt. It ensures that illustrations show authentic teacher facilitation, child participation, questioning, feedback, inclusion and classroom routines aligned with BCube EduOS™.

1. Teaching Philosophy

Objective: Standardize the teaching behaviours represented in every BCube page.

- Teacher as facilitator
- Child agency
- Learning through interaction
- Positive reinforcement
- Safe classroom culture

Compiler Output

Produces structured teaching metadata including facilitation strategy, questioning cues, observation focus and feedback guidance.

Acceptance Criteria

Every generated page depicts meaningful teacher-child interaction that supports the learning objective without overwhelming the learner.

2. BCube Learning Experience Model™

Objective: Standardize the teaching behaviours represented in every BCube page.

- Wonder
- Explore
- Talk
- Practice
- Create
- Reflect
- Celebrate learning

Compiler Output

Produces structured teaching metadata including facilitation strategy, questioning cues, observation focus and feedback guidance.

Acceptance Criteria

Every generated page depicts meaningful teacher-child interaction that supports the learning objective without overwhelming the learner.

3. Teacher Interaction Standards

Objective: Standardize the teaching behaviours represented in every BCube page.

- Open-ended questions
- Scaffolding
- Model then guide
- Observe before intervening
- Encourage peer learning

Compiler Output

Produces structured teaching metadata including facilitation strategy, questioning cues, observation focus and feedback guidance.

Acceptance Criteria

Every generated page depicts meaningful teacher-child interaction that supports the learning objective without overwhelming the learner.

4. Classroom Facilitation

Objective: Standardize the teaching behaviours represented in every BCube page.

- Circle time
- Small groups
- Independent activity
- Transitions
- Routine consistency

Compiler Output

Produces structured teaching metadata including facilitation strategy, questioning cues, observation focus and feedback guidance.

Acceptance Criteria

Every generated page depicts meaningful teacher-child interaction that supports the learning objective without overwhelming the learner.

5. Questioning Framework

Objective: Standardize the teaching behaviours represented in every BCube page.

- Recall
- Observe
- Compare
- Predict
- Explain
- Reflect

Compiler Output

Produces structured teaching metadata including facilitation strategy, questioning cues, observation focus and feedback guidance.

Acceptance Criteria

Every generated page depicts meaningful teacher-child interaction that supports the learning objective without overwhelming the learner.

6. Feedback Standards

Objective: Standardize the teaching behaviours represented in every BCube page.

- Immediate feedback
- Specific praise
- Growth-focused language
- Constructive correction
- Celebrate effort

Compiler Output

Produces structured teaching metadata including facilitation strategy, questioning cues, observation focus and feedback guidance.

Acceptance Criteria

Every generated page depicts meaningful teacher-child interaction that supports the learning objective without overwhelming the learner.

7. Inclusive Practice

Objective: Standardize the teaching behaviours represented in every BCube page.

- Multiple participation methods
- Flexible support
- Accessible communication
- Respectful representation

Compiler Output

Produces structured teaching metadata including facilitation strategy, questioning cues, observation focus and feedback guidance.

Acceptance Criteria

Every generated page depicts meaningful teacher-child interaction that supports the learning objective without overwhelming the learner.

8. Observation & Assessment

Objective: Standardize the teaching behaviours represented in every BCube page.

- Anecdotal observation
- Learning evidence
- Prompt adaptation
- Next-step planning

Compiler Output

Produces structured teaching metadata including facilitation strategy, questioning cues, observation focus and feedback guidance.

Acceptance Criteria

Every generated page depicts meaningful teacher-child interaction that supports the learning objective without overwhelming the learner.

9. Teacher Metadata

Objective: Standardize the teaching behaviours represented in every BCube page.

- Teaching strategy
- Interaction type
- Question prompts
- Observation focus
- Feedback cue

Compiler Output

Produces structured teaching metadata including facilitation strategy, questioning cues, observation focus and feedback guidance.

Acceptance Criteria

Every generated page depicts meaningful teacher-child interaction that supports the learning objective without overwhelming the learner.

10. Teaching Validation

Objective: Standardize the teaching behaviours represented in every BCube page.

- Teacher visible
- Learning interaction evident
- Questioning included
- Child engagement authentic
- EduOS compliant

Compiler Output

Produces structured teaching metadata including facilitation strategy, questioning cues, observation focus and feedback guidance.

Acceptance Criteria

Every generated page depicts meaningful teacher-child interaction that supports the learning objective without overwhelming the learner.

Teaching Metadata Schema

Field	Purpose	Required
Teaching Strategy	Instructional approach	Yes

Field	Purpose	Required
Teacher Action	Primary facilitation	Yes
Question Prompt	Conversation starter	Yes
Observation Focus	Evidence to notice	Yes
Feedback Cue	Reinforcement guidance	Optional
Group Format	Whole/small/individual	Optional
Home Link	Carry-over activity	Optional

Negative Teaching Rules

- No passive teacher standing without interaction
- No lecture-style instruction for preschool
- No punitive or fear-based language
- No unsafe classroom behaviour
- No child excluded from participation
- No activity without teacher facilitation cues

Phase 5 Deliverables

- Teaching Engine™ specification
- Learning Experience Model integration
- Questioning Framework
- Teacher Interaction Standards
- Observation & Feedback Guide
- Teaching Metadata Schema
- Teaching Validation Checklist
- Compiler-ready Teaching Block

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Master_Part_7.docx`

Purpose

This section defines the Teaching Engine™ as the instructional orchestration layer. It standardizes teacher facilitation, questioning, scaffolding, classroom interaction, feedback, observation, classroom culture and instructional metadata.

TE-FAC-001

Requirement: Every learning activity SHALL include a defined teacher facilitation strategy.

Rationale: Promotes consistent, evidence-based classroom practice.

Inputs: Educational metadata, activity type, learner profile.

Outputs: Teaching strategy, questioning prompts, facilitation metadata.

Validation: Check instructional alignment, questioning quality and evidence collection.

Compiler Impact: Produces the Teaching Engine™ block for the final production prompt.

TE-QUE-001

Requirement: Teachers SHOULD use open-ended questioning before direct explanation.

Rationale: Promotes consistent, evidence-based classroom practice.

Inputs: Educational metadata, activity type, learner profile.

Outputs: Teaching strategy, questioning prompts, facilitation metadata.

Validation: Check instructional alignment, questioning quality and evidence collection.

Compiler Impact: Produces the Teaching Engine™ block for the final production prompt.

TE-SCF-001

Requirement: Scaffolding MUST progress from modelling to guided practice to independence.

Rationale: Promotes consistent, evidence-based classroom practice.

Inputs: Educational metadata, activity type, learner profile.

Outputs: Teaching strategy, questioning prompts, facilitation metadata.

Validation: Check instructional alignment, questioning quality and evidence collection.

Compiler Impact: Produces the Teaching Engine™ block for the final production prompt.

TE-INT-001

Requirement: Teacher-child interactions SHALL reinforce the primary learning objective.

Rationale: Promotes consistent, evidence-based classroom practice.

Inputs: Educational metadata, activity type, learner profile.

Outputs: Teaching strategy, questioning prompts, facilitation metadata.

Validation: Check instructional alignment, questioning quality and evidence collection.

Compiler Impact: Produces the Teaching Engine™ block for the final production prompt.

TE-OBS-001

Requirement: Teachers MUST observe and document evidence of learning.

Rationale: Promotes consistent, evidence-based classroom practice.

Inputs: Educational metadata, activity type, learner profile.

Outputs: Teaching strategy, questioning prompts, facilitation metadata.

Validation: Check instructional alignment, questioning quality and evidence collection.

Compiler Impact: Produces the Teaching Engine™ block for the final production prompt.

TE-FDB-001

Requirement: Feedback SHALL be specific, positive and growth-oriented.

Rationale: Promotes consistent, evidence-based classroom practice.

Inputs: Educational metadata, activity type, learner profile.

Outputs: Teaching strategy, questioning prompts, facilitation metadata.

Validation: Check instructional alignment, questioning quality and evidence collection.

Compiler Impact: Produces the Teaching Engine™ block for the final production prompt.

TE-CLS-001

Requirement: Classroom routines SHOULD promote safety, inclusion and engagement.

Rationale: Promotes consistent, evidence-based classroom practice.

Inputs: Educational metadata, activity type, learner profile.

Outputs: Teaching strategy, questioning prompts, facilitation metadata.

Validation: Check instructional alignment, questioning quality and evidence collection.

Compiler Impact: Produces the Teaching Engine™ block for the final production prompt.

TE-GRP-001

Requirement: Activities MAY support whole-class, small-group or individual learning.

Rationale: Promotes consistent, evidence-based classroom practice.

Inputs: Educational metadata, activity type, learner profile.

Outputs: Teaching strategy, questioning prompts, facilitation metadata.

Validation: Check instructional alignment, questioning quality and evidence collection.

Compiler Impact: Produces the Teaching Engine™ block for the final production prompt.

TE-MTD-001

Requirement: Teaching metadata SHALL be complete before prompt compilation.

Rationale: Promotes consistent, evidence-based classroom practice.

Inputs: Educational metadata, activity type, learner profile.

Outputs: Teaching strategy, questioning prompts, facilitation metadata.

Validation: Check instructional alignment, questioning quality and evidence collection.

Compiler Impact: Produces the Teaching Engine™ block for the final production prompt.

TE-VAL-001

Requirement: Teaching validation MUST pass before QA approval.

Rationale: Promotes consistent, evidence-based classroom practice.

Inputs: Educational metadata, activity type, learner profile.

Outputs: Teaching strategy, questioning prompts, facilitation metadata.

Validation: Check instructional alignment, questioning quality and evidence collection.

Compiler Impact: Produces the Teaching Engine™ block for the final production prompt.

Teaching Metadata Matrix

Field	Purpose	Required
Teaching Strategy	Instructional approach	Yes
Question Prompt	Teacher dialogue	Yes
Observation Focus	Evidence collection	Yes
Feedback Cue	Positive reinforcement	Yes
Grouping	Class organization	Optional
Resources	Teaching materials	Optional
Reflection	Lesson closure	Optional

Teaching Anti-Patterns

- Teacher dominates instead of facilitating
- Questions with only yes/no answers
- No opportunity for child exploration
- Feedback focused only on correctness
- Lack of observation evidence
- Unsafe or exclusionary classroom practice

Cross References

Related rules: EE-ACT-001, PP-HME-001, QA-TEA-001, PC-COMP-001.

Expansion Roadmap

Future revisions will expand each Teaching Engine™ rule with classroom scenarios, dialogue examples, scaffolding matrices, observation rubrics, implementation guidance, prompt fragments and automated validation logic.

Phase 6 - Parent Partnership Engine

Phase 6 - Parent Partnership Engine

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Purpose

The Parent Partnership Engine™ ensures every BCube learning experience extends beyond the classroom. It standardizes parent engagement, home learning, conversation prompts, reinforcement activities, and communication practices so families become active partners in children's development.

1. Parent Partnership Philosophy

Objective: Standardize parent engagement across every BCube publication.

- Parents as learning partners
- Positive home-school collaboration
- Short, practical home support
- Respect family diversity
- Celebrate progress

Compiler Output

Produces parent-facing metadata including conversation prompts, home reinforcement, family activity ideas and communication guidance.

Acceptance Criteria

Every page includes a realistic, optional home extension that reinforces classroom learning without creating unnecessary burden for families.

2. Home Learning Framework

Objective: Standardize parent engagement across every BCube publication.

- 5–10 minute activities
- Everyday object usage
- Play-based reinforcement
- Language-rich interactions
- Routine integration

Compiler Output

Produces parent-facing metadata including conversation prompts, home reinforcement, family activity ideas and communication guidance.

Acceptance Criteria

Every page includes a realistic, optional home extension that reinforces classroom learning without creating unnecessary burden for families.

3. Conversation Prompts

Objective: Standardize parent engagement across every BCube publication.

- Open-ended questions
- Observation prompts
- Reflection prompts
- Vocabulary reinforcement
- Emotional check-ins

Compiler Output

Produces parent-facing metadata including conversation prompts, home reinforcement, family activity ideas and communication guidance.

Acceptance Criteria

Every page includes a realistic, optional home extension that reinforces classroom learning without creating unnecessary burden for families.

4. Family Engagement

Objective: Standardize parent engagement across every BCube publication.

- Shared reading
- Family games
- Creative projects
- Real-life exploration
- Celebrating achievements

Compiler Output

Produces parent-facing metadata including conversation prompts, home reinforcement, family activity ideas and communication guidance.

Acceptance Criteria

Every page includes a realistic, optional home extension that reinforces classroom learning without creating unnecessary burden for families.

5. Reinforcement Strategy

Objective: Standardize parent engagement across every BCube publication.

- Positive praise
- Growth mindset language
- Repetition without pressure
- Choice-based participation

Compiler Output

Produces parent-facing metadata including conversation prompts, home reinforcement, family activity ideas and communication guidance.

Acceptance Criteria

Every page includes a realistic, optional home extension that reinforces classroom learning without creating unnecessary burden for families.

6. Parent Communication

Objective: Standardize parent engagement across every BCube publication.

- Simple language
- Clear expectations
- Progress updates
- Next steps
- Home-school consistency

Compiler Output

Produces parent-facing metadata including conversation prompts, home reinforcement, family activity ideas and communication guidance.

Acceptance Criteria

Every page includes a realistic, optional home extension that reinforces classroom learning without creating unnecessary burden for families.

7. Inclusive Family Practices

Objective: Standardize parent engagement across every BCube publication.

- Flexible participation
- Different family structures
- Accessible activities
- Low-cost resources

Compiler Output

Produces parent-facing metadata including conversation prompts, home reinforcement, family activity ideas and communication guidance.

Acceptance Criteria

Every page includes a realistic, optional home extension that reinforces classroom learning without creating unnecessary burden for families.

8. Parent Metadata

Objective: Standardize parent engagement across every BCube publication.

- Home activity
- Conversation starter
- Materials
- Estimated time
- Learning outcome

Compiler Output

Produces parent-facing metadata including conversation prompts, home reinforcement, family activity ideas and communication guidance.

Acceptance Criteria

Every page includes a realistic, optional home extension that reinforces classroom learning without creating unnecessary burden for families.

9. Validation Rules

Objective: Standardize parent engagement across every BCube publication.

- Home activity supports objective
- No excessive preparation
- Age-appropriate
- Family friendly
- Optional extension

Compiler Output

Produces parent-facing metadata including conversation prompts, home reinforcement, family activity ideas and communication guidance.

Acceptance Criteria

Every page includes a realistic, optional home extension that reinforces classroom learning without creating unnecessary burden for families.

10. Governance

Objective: Standardize parent engagement across every BCube publication.

- Review cycle
- Evidence collection
- Feedback loop
- Continuous improvement
- EduOS alignment

Compiler Output

Produces parent-facing metadata including conversation prompts, home reinforcement, family activity ideas and communication guidance.

Acceptance Criteria

Every page includes a realistic, optional home extension that reinforces classroom learning without creating unnecessary burden for families.

Parent Metadata Schema

Field	Purpose	Required
Conversation Prompt	Family discussion	Yes

Field	Purpose	Required
Home Activity	Reinforcement	Yes
Estimated Time	Practical planning	Yes
Materials	Common household items	Optional
Expected Outcome	Learning evidence	Yes
Extension	Optional challenge	Optional

Negative Parent Rules

- No homework requiring expensive materials
- No activities exceeding recommended duration
- No parent-only completion
- No assessment pressure at home
- No inaccessible language
- No unrealistic family expectations

Phase 6 Deliverables

- Parent Partnership Engine™ specification
- Home Learning Framework
- Conversation Prompt Library
- Family Engagement Standards
- Parent Metadata Schema
- Parent Validation Checklist
- Compiler-ready Parent Block

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Master_Part_8.docx`

Purpose

The Parent Partnership Engine™ defines the standards for extending learning beyond the classroom. It governs home-learning activities, family engagement, parent communication, reinforcement strategies, accessibility, and metadata that supports meaningful home-school collaboration.

PP-HME-001

Requirement: Every learning unit SHOULD include an optional home-learning activity.

Rationale: Strengthens continuity between classroom learning and the home environment.

Inputs: Educational objective, learner profile, activity metadata.

Outputs: Conversation prompts, home activities, reinforcement metadata.

Validation: Verify clarity, feasibility, inclusiveness and alignment with learning objectives.

Compiler Impact: Produces the Parent Partnership Engine™ block for the compiled production prompt.

PP-COM-001

Requirement: Parent communication SHALL use clear, encouraging and jargon-free language.

Rationale: Strengthens continuity between classroom learning and the home environment.

Inputs: Educational objective, learner profile, activity metadata.

Outputs: Conversation prompts, home activities, reinforcement metadata.

Validation: Verify clarity, feasibility, inclusiveness and alignment with learning objectives.

Compiler Impact: Produces the Parent Partnership Engine™ block for the compiled production prompt.

PP-CON-001

Requirement: Conversation prompts MUST reinforce the classroom learning objective.

Rationale: Strengthens continuity between classroom learning and the home environment.

Inputs: Educational objective, learner profile, activity metadata.

Outputs: Conversation prompts, home activities, reinforcement metadata.

Validation: Verify clarity, feasibility, inclusiveness and alignment with learning objectives.

Compiler Impact: Produces the Parent Partnership Engine™ block for the compiled production prompt.

PP-ENG-001

Requirement: Family engagement activities SHOULD be practical and inclusive.

Rationale: Strengthens continuity between classroom learning and the home environment.

Inputs: Educational objective, learner profile, activity metadata.

Outputs: Conversation prompts, home activities, reinforcement metadata.

Validation: Verify clarity, feasibility, inclusiveness and alignment with learning objectives.

Compiler Impact: Produces the Parent Partnership Engine™ block for the compiled production prompt.

PP-RES-001

Requirement: Home activities MUST rely on commonly available materials whenever possible.

Rationale: Strengthens continuity between classroom learning and the home environment.

Inputs: Educational objective, learner profile, activity metadata.

Outputs: Conversation prompts, home activities, reinforcement metadata.

Validation: Verify clarity, feasibility, inclusiveness and alignment with learning objectives.

Compiler Impact: Produces the Parent Partnership Engine™ block for the compiled production prompt.

PP-TIM-001

Requirement: Recommended activity duration SHALL match the developmental level.

Rationale: Strengthens continuity between classroom learning and the home environment.

Inputs: Educational objective, learner profile, activity metadata.

Outputs: Conversation prompts, home activities, reinforcement metadata.

Validation: Verify clarity, feasibility, inclusiveness and alignment with learning objectives.

Compiler Impact: Produces the Parent Partnership Engine™ block for the compiled production prompt.

PP-ACC-001

Requirement: Home guidance MUST be accessible to diverse family structures and contexts.

Rationale: Strengthens continuity between classroom learning and the home environment.

Inputs: Educational objective, learner profile, activity metadata.

Outputs: Conversation prompts, home activities, reinforcement metadata.

Validation: Verify clarity, feasibility, inclusiveness and alignment with learning objectives.

Compiler Impact: Produces the Parent Partnership Engine™ block for the compiled production prompt.

PP-MTD-001

Requirement: Parent metadata SHALL be complete before compilation.

Rationale: Strengthens continuity between classroom learning and the home environment.

Inputs: Educational objective, learner profile, activity metadata.

Outputs: Conversation prompts, home activities, reinforcement metadata.

Validation: Verify clarity, feasibility, inclusiveness and alignment with learning objectives.

Compiler Impact: Produces the Parent Partnership Engine™ block for the compiled production prompt.

PP-EXT-001

Requirement: Optional extension activities MAY be provided for deeper exploration.

Rationale: Strengthens continuity between classroom learning and the home environment.

Inputs: Educational objective, learner profile, activity metadata.

Outputs: Conversation prompts, home activities, reinforcement metadata.

Validation: Verify clarity, feasibility, inclusiveness and alignment with learning objectives.

Compiler Impact: Produces the Parent Partnership Engine™ block for the compiled production prompt.

PP-VAL-001

Requirement: Parent Partnership validation MUST pass before QA approval.

Rationale: Strengthens continuity between classroom learning and the home environment.

Inputs: Educational objective, learner profile, activity metadata.

Outputs: Conversation prompts, home activities, reinforcement metadata.

Validation: Verify clarity, feasibility, inclusiveness and alignment with learning objectives.

Compiler Impact: Produces the Parent Partnership Engine™ block for the compiled production prompt.

Parent Metadata Matrix

Field	Purpose	Required
Conversation Prompt	Guide family discussion	Yes
Home Activity	Reinforce classroom learning	Yes
Estimated Time	Family planning	Yes
Materials	Household resources	Optional
Expected Outcome	Observable reinforcement	Yes
Extension Activity	Additional practice	Optional
Parent Notes	Supportive guidance	Optional

Parent Partnership Anti-Patterns

- Homework requiring expensive resources
- Activities exceeding recommended duration
- Instructions written in technical language
- Assuming a single family structure
- Home tasks disconnected from classroom objectives
- Assessment-focused rather than encouragement-focused guidance

Cross References

Related rules: EE-TRN-001, TE-FDB-001, QA-PAR-001, PC-COMP-001.

Expansion Roadmap

Future revisions will add detailed family engagement frameworks, multilingual communication guidance, home-learning templates, accessibility recommendations, validation algorithms and reusable prompt fragments.

Phase 7 - Illustration Engine

Phase 7 - Illustration Engine

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Purpose

The Illustration Engine™ converts educational intent into consistent visual scenes. It defines the artistic language, rendering style, character placement, environments, lighting, composition, colour harmony and technical illustration rules required for every BCube publication.

1. Illustration Philosophy

Objective: Define mandatory illustration rules used by the Prompt Compiler™.

- Illustration teaches, not decorates
- Emotion supports learning
- Age-appropriate simplicity
- Consistent premium visual identity
- Educational realism with warmth

Compiler Output

Outputs structured illustration metadata describing scene, rendering style, composition, characters, environment, colour palette and technical generation constraints.

Acceptance Criteria

Illustrations are visually consistent, educationally purposeful, technically printable and aligned with BCube EduOS™.

2. Artistic Style Standards

Objective: Define mandatory illustration rules used by the Prompt Compiler™.

- Clean vector-like finish
- Soft pastel palette
- Rounded friendly forms
- Thick readable outlines
- Print-first rendering
- Consistent proportions

Compiler Output

Outputs structured illustration metadata describing scene, rendering style, composition, characters, environment, colour palette and technical generation constraints.

Acceptance Criteria

Illustrations are visually consistent, educationally purposeful, technically printable and aligned with BCube EduOS™.

3. Scene Composition

Objective: Define mandatory illustration rules used by the Prompt Compiler™.

- Single dominant learning focus
- Balanced foreground/midground/background
- Clear activity space
- Natural classroom storytelling
- Protected text zones

Compiler Output

Outputs structured illustration metadata describing scene, rendering style, composition, characters, environment, colour palette and technical generation constraints.

Acceptance Criteria

Illustrations are visually consistent, educationally purposeful, technically printable and aligned with BCube EduOS™.

4. Character Placement

Objective: Define mandatory illustration rules used by the Prompt Compiler™.

- Teacher facilitates rather than dominates
- Star reinforces learning
- Children actively engaged
- Inclusive representation
- Natural eye contact

Compiler Output

Outputs structured illustration metadata describing scene, rendering style, composition, characters, environment, colour palette and technical generation constraints.

Acceptance Criteria

Illustrations are visually consistent, educationally purposeful, technically printable and aligned with BCube EduOS™.

5. Environment Library

Objective: Define mandatory illustration rules used by the Prompt Compiler™.

- Classroom
- Home
- Playground
- Nature
- Community
- STEM corner
- Art corner

Compiler Output

Outputs structured illustration metadata describing scene, rendering style, composition, characters, environment, colour palette and technical generation constraints.

Acceptance Criteria

Illustrations are visually consistent, educationally purposeful, technically printable and aligned with BCube EduOS™.

6. Lighting & Colour

Objective: Define mandatory illustration rules used by the Prompt Compiler™.

- Soft daylight
- Consistent shadows
- Warm colour harmony
- High contrast for learning objects
- No harsh effects

Compiler Output

Outputs structured illustration metadata describing scene, rendering style, composition, characters, environment, colour palette and technical generation constraints.

Acceptance Criteria

Illustrations are visually consistent, educationally purposeful, technically printable and aligned with BCube EduOS™.

7. Object & Prop Standards

Objective: Define mandatory illustration rules used by the Prompt Compiler™.

- Age-appropriate furniture
- Recognizable objects
- Consistent scale
- Minimal clutter
- Safe classroom equipment

Compiler Output

Outputs structured illustration metadata describing scene, rendering style, composition, characters, environment, colour palette and technical generation constraints.

Acceptance Criteria

Illustrations are visually consistent, educationally purposeful, technically printable and aligned with BCube EduOS™.

8. AI Illustration Rules

Objective: Define mandatory illustration rules used by the Prompt Compiler™.

- No embedded text

- No watermarks
- No anatomy errors
- No cropped key elements
- Editable composition
- Protected activity zones

Compiler Output

Outputs structured illustration metadata describing scene, rendering style, composition, characters, environment, colour palette and technical generation constraints.

Acceptance Criteria

Illustrations are visually consistent, educationally purposeful, technically printable and aligned with BCube EduOS™.

9. Illustration Metadata

Objective: Define mandatory illustration rules used by the Prompt Compiler™.

- Scene type
- Camera angle
- Perspective
- Mood
- Palette
- Characters
- Props
- Environment

Compiler Output

Outputs structured illustration metadata describing scene, rendering style, composition, characters, environment, colour palette and technical generation constraints.

Acceptance Criteria

Illustrations are visually consistent, educationally purposeful, technically printable and aligned with BCube EduOS™.

10. Illustration Validation

Objective: Define mandatory illustration rules used by the Prompt Compiler™.

- Educational clarity
- Visual consistency
- Brand alignment
- Accessibility
- Print quality
- Prompt Engine compliance

Compiler Output

Outputs structured illustration metadata describing scene, rendering style, composition, characters, environment, colour palette and technical generation constraints.

Acceptance Criteria

Illustrations are visually consistent, educationally purposeful, technically printable and aligned with BCube EduOS™.

Illustration Metadata Schema

Field	Purpose	Required
Scene Type	Context	Yes
Environment	Location	Yes
Characters	Teacher/Star/Children	Yes
Primary Learning Object	Focus	Yes
Colour Palette	Visual consistency	Yes
Camera Angle	Composition	Optional
Mood	Emotional tone	Optional
Props	Supporting objects	Optional

Negative Illustration Rules

- No distorted anatomy or extra fingers
- No random text or logos
- No overcrowded backgrounds
- No unsafe classroom scenes
- No inconsistent Star mascot design
- No photorealistic style for standard workbook interiors
- No hidden learning objects
- No blocked activity areas

Phase 7 Deliverables

- Illustration Engine™ specification
- Artistic Style Guide
- Environment Library
- Scene Composition Rules
- Illustration Metadata Schema
- Illustration Validation Checklist
- Compiler-ready Illustration Block

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Master_Part_9.docx`

Purpose

The Illustration Engine™ defines the technical and artistic execution standards for all BCube visual assets. It governs rendering style, scene composition, environment design, colour systems, lighting, perspective, prop usage, visual continuity and illustration metadata required for consistent AI-assisted production.

IE-REN-001

Requirement: Every illustration SHALL use the approved BCube rendering style and visual identity.

Rationale: Ensure premium-quality, educationally effective and visually consistent illustrations.

Inputs: Publishing, Design, Visual Grammar and Educational Engine metadata.

Outputs: Scene specification, rendering parameters, environment metadata and illustration directives.

Validation: Check composition, quality, accessibility, continuity and print suitability.

Compiler Impact: Produces the Illustration Engine™ block for the BCube Gold Production Prompt™.

IE-SCN-001

Requirement: Each page MUST communicate one coherent educational scene.

Rationale: Ensure premium-quality, educationally effective and visually consistent illustrations.

Inputs: Publishing, Design, Visual Grammar and Educational Engine metadata.

Outputs: Scene specification, rendering parameters, environment metadata and illustration directives.

Validation: Check composition, quality, accessibility, continuity and print suitability.

Compiler Impact: Produces the Illustration Engine™ block for the BCube Gold Production Prompt™.

IE-ENV-001

Requirement: Background environments SHALL reinforce the learning objective without distraction.

Rationale: Ensure premium-quality, educationally effective and visually consistent illustrations.

Inputs: Publishing, Design, Visual Grammar and Educational Engine metadata.

Outputs: Scene specification, rendering parameters, environment metadata and illustration directives.

Validation: Check composition, quality, accessibility, continuity and print suitability.

Compiler Impact: Produces the Illustration Engine™ block for the BCube Gold Production Prompt™.

IE-LGT-001

Requirement: Lighting MUST maximize readability and maintain a warm, child-friendly atmosphere.

Rationale: Ensure premium-quality, educationally effective and visually consistent illustrations.

Inputs: Publishing, Design, Visual Grammar and Educational Engine metadata.

Outputs: Scene specification, rendering parameters, environment metadata and illustration directives.

Validation: Check composition, quality, accessibility, continuity and print suitability.

Compiler Impact: Produces the Illustration Engine™ block for the BCube Gold Production Prompt™.

IE-PER-001

Requirement: Perspective SHALL remain age-appropriate and visually stable.

Rationale: Ensure premium-quality, educationally effective and visually consistent illustrations.

Inputs: Publishing, Design, Visual Grammar and Educational Engine metadata.

Outputs: Scene specification, rendering parameters, environment metadata and illustration directives.

Validation: Check composition, quality, accessibility, continuity and print suitability.

Compiler Impact: Produces the Illustration Engine™ block for the BCube Gold Production Prompt™.

IE-PRO-001

Requirement: Props MUST support learning and never compete with instructional content.

Rationale: Ensure premium-quality, educationally effective and visually consistent illustrations.

Inputs: Publishing, Design, Visual Grammar and Educational Engine metadata.

Outputs: Scene specification, rendering parameters, environment metadata and illustration directives.

Validation: Check composition, quality, accessibility, continuity and print suitability.

Compiler Impact: Produces the Illustration Engine™ block for the BCube Gold Production Prompt™.

IE-COL-001

Requirement: Illustrations SHALL use the approved BCube colour system and accessibility guidelines.

Rationale: Ensure premium-quality, educationally effective and visually consistent illustrations.

Inputs: Publishing, Design, Visual Grammar and Educational Engine metadata.

Outputs: Scene specification, rendering parameters, environment metadata and illustration directives.

Validation: Check composition, quality, accessibility, continuity and print suitability.

Compiler Impact: Produces the Illustration Engine™ block for the BCube Gold Production Prompt™.

IE-CNT-001

Requirement: Recurring scenes and assets MUST maintain continuity across the series.

Rationale: Ensure premium-quality, educationally effective and visually consistent illustrations.

Inputs: Publishing, Design, Visual Grammar and Educational Engine metadata.

Outputs: Scene specification, rendering parameters, environment metadata and illustration directives.

Validation: Check composition, quality, accessibility, continuity and print suitability.

Compiler Impact: Produces the Illustration Engine™ block for the BCube Gold Production Prompt™.

IE-MTD-001

Requirement: Illustration metadata SHALL be complete before compilation.

Rationale: Ensure premium-quality, educationally effective and visually consistent illustrations.

Inputs: Publishing, Design, Visual Grammar and Educational Engine metadata.

Outputs: Scene specification, rendering parameters, environment metadata and illustration directives.

Validation: Check composition, quality, accessibility, continuity and print suitability.

Compiler Impact: Produces the Illustration Engine™ block for the BCube Gold Production Prompt™.

IE-VAL-001

Requirement: Illustration validation MUST pass before Character Engine and QA execution.

Rationale: Ensure premium-quality, educationally effective and visually consistent illustrations.

Inputs: Publishing, Design, Visual Grammar and Educational Engine metadata.

Outputs: Scene specification, rendering parameters, environment metadata and illustration directives.

Validation: Check composition, quality, accessibility, continuity and print suitability.

Compiler Impact: Produces the Illustration Engine™ block for the BCube Gold Production Prompt™.

Illustration Metadata Matrix

Field	Purpose	Required
Scene Type	Educational context	Yes
Environment	Learning setting	Yes
Rendering Style	Visual consistency	Yes
Primary Characters	Scene focus	Yes
Supporting Props	Learning support	Optional
Lighting	Readability and mood	Yes
Perspective	Composition stability	Yes
Colour Profile	Brand consistency	Yes
Continuity Tag	Series consistency	Optional

Illustration Anti-Patterns

- Photorealistic rendering in preschool workbooks
- Overcrowded backgrounds
- Irrelevant decorative props
- Inconsistent Star mascot appearance
- Low-resolution or blurred artwork
- Text embedded incorrectly within illustrations
- Unsafe or confusing classroom scenes

Cross References

Related rules: VG-SCN-001, CE-STR-001, QA-ILL-001, PC-ASM-001.

Expansion Roadmap

Future revisions will extend each Illustration Engine™ rule with detailed rendering specifications, environment libraries, prop catalogs, lighting profiles, perspective templates, validation algorithms, prompt fragments and reference illustrations.

Phase 8 - Character Engine

Phase 8 - Character Engine

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Purpose

The Character Engine™ governs every recurring BCube character. It standardizes appearance, proportions, expressions, poses, behaviour, continuity, diversity, accessibility and interaction rules so characters remain consistent across books, grades and media.

1. Character Philosophy

Objective: Standardize recurring character behaviour and visual identity.

- Characters support learning
- Warm and trustworthy personalities
- Age-appropriate design
- Consistency across all products
- Inclusive representation

Compiler Output

Produces structured character metadata including identity, pose, expression, interaction, wardrobe and continuity rules.

Acceptance Criteria

Characters remain instantly recognizable, emotionally appropriate and educationally supportive across every BCube publication.

2. Star Mascot Standard

Objective: Standardize recurring character behaviour and visual identity.

- Learning guide, not distraction
- Encouraging expressions
- Supports objectives
- Consistent colours and proportions
- Positive gestures

Compiler Output

Produces structured character metadata including identity, pose, expression, interaction, wardrobe and continuity rules.

Acceptance Criteria

Characters remain instantly recognizable, emotionally appropriate and educationally supportive across every BCube publication.

3. Teacher Character Standards

Objective: Standardize recurring character behaviour and visual identity.

- Facilitator role
- Professional yet approachable
- Inclusive interactions
- Natural classroom positioning
- Positive body language

Compiler Output

Produces structured character metadata including identity, pose, expression, interaction, wardrobe and continuity rules.

Acceptance Criteria

Characters remain instantly recognizable, emotionally appropriate and educationally supportive across every BCube publication.

4. Child Character Library

Objective: Standardize recurring character behaviour and visual identity.

- Diverse representation
- Developmentally appropriate proportions
- Active participation
- Natural emotions
- Respectful inclusion

Compiler Output

Produces structured character metadata including identity, pose, expression, interaction, wardrobe and continuity rules.

Acceptance Criteria

Characters remain instantly recognizable, emotionally appropriate and educationally supportive across every BCube publication.

5. Family & Community Characters

Objective: Standardize recurring character behaviour and visual identity.

- Parents
- Grandparents
- Community helpers
- Visitors
- Multiple family structures

Compiler Output

Produces structured character metadata including identity, pose, expression, interaction, wardrobe and continuity rules.

Acceptance Criteria

Characters remain instantly recognizable, emotionally appropriate and educationally supportive across every BCube publication.

6. Pose & Expression System

Objective: Standardize recurring character behaviour and visual identity.

- Happy
- Curious
- Thinking
- Listening
- Helping
- Celebrating
- Standard pose library
- Gesture consistency

Compiler Output

Produces structured character metadata including identity, pose, expression, interaction, wardrobe and continuity rules.

Acceptance Criteria

Characters remain instantly recognizable, emotionally appropriate and educationally supportive across every BCube publication.

7. Continuity Rules

Objective: Standardize recurring character behaviour and visual identity.

- Consistent clothing themes
- Character scale
- Recurring visual identifiers
- Scene-to-scene consistency

Compiler Output

Produces structured character metadata including identity, pose, expression, interaction, wardrobe and continuity rules.

Acceptance Criteria

Characters remain instantly recognizable, emotionally appropriate and educationally supportive across every BCube publication.

8. Interaction Rules

Objective: Standardize recurring character behaviour and visual identity.

- Teacher-child
- Child-child
- Star-child

- Family-child
- Positive collaboration
- Safe behaviour

Compiler Output

Produces structured character metadata including identity, pose, expression, interaction, wardrobe and continuity rules.

Acceptance Criteria

Characters remain instantly recognizable, emotionally appropriate and educationally supportive across every BCube publication.

9. Character Metadata

Objective: Standardize recurring character behaviour and visual identity.

- Character ID
- Role
- Expression
- Pose
- Interaction type
- Wardrobe
- Emotion
- Scene importance

Compiler Output

Produces structured character metadata including identity, pose, expression, interaction, wardrobe and continuity rules.

Acceptance Criteria

Characters remain instantly recognizable, emotionally appropriate and educationally supportive across every BCube publication.

10. Character Validation

Objective: Standardize recurring character behaviour and visual identity.

- Identity preserved
- Expressions appropriate
- Inclusive representation
- No conflicting designs
- Brand compliant
- EduOS aligned

Compiler Output

Produces structured character metadata including identity, pose, expression, interaction, wardrobe and continuity rules.

Acceptance Criteria

Characters remain instantly recognizable, emotionally appropriate and educationally supportive across every BCube publication.

Character Metadata Schema

Field	Purpose	Required
Character ID	Unique reference	Yes
Role	Star/Teacher/Child/Parent	Yes
Expression	Emotion	Yes
Pose	Body language	Yes
Interaction	Relationship	Yes
Wardrobe	Continuity	Optional
Scene Priority	Primary/Secondary	Yes
Accessibility Notes	Inclusive representation	Optional

Negative Character Rules

- No inconsistent appearance between pages
- No exaggerated or frightening expressions
- No unsafe behaviours
- No stereotypical representation
- No copyrighted external characters
- No random wardrobe changes without narrative reason
- No characters blocking learning activities

Phase 8 Deliverables

- Character Engine™ specification
- Star Mascot Standard
- Teacher & Child Character Library
- Pose & Expression Guide
- Interaction Standards
- Character Metadata Schema
- Character Validation Checklist
- Compiler-ready Character Block

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Master_Part_10.docx`

Purpose

The Character Engine™ establishes the canonical standards for every recurring BCube character. It defines identity, proportions, expressions, poses, interaction rules, continuity, inclusivity, wardrobe, accessibility and metadata so all characters remain consistent across books and media.

CE-STR-001

Requirement: Star mascot SHALL remain visually consistent across all publications.

Rationale: Maintain recognisable, trustworthy and educationally effective characters.

Inputs: Illustration metadata, curriculum context, page objective and continuity profile.

Outputs: Character specification, pose, expression and interaction metadata.

Validation: Verify identity, continuity, accessibility and educational appropriateness.

Compiler Impact: Produces the Character Engine™ block for the BCube Gold Production Prompt™.

CE-IDN-001

Requirement: Every recurring character MUST have a unique canonical identity.

Rationale: Maintain recognisable, trustworthy and educationally effective characters.

Inputs: Illustration metadata, curriculum context, page objective and continuity profile.

Outputs: Character specification, pose, expression and interaction metadata.

Validation: Verify identity, continuity, accessibility and educational appropriateness.

Compiler Impact: Produces the Character Engine™ block for the BCube Gold Production Prompt™.

CE-EXP-001

Requirement: Expressions SHALL support the intended learning emotion.

Rationale: Maintain recognisable, trustworthy and educationally effective characters.

Inputs: Illustration metadata, curriculum context, page objective and continuity profile.

Outputs: Character specification, pose, expression and interaction metadata.

Validation: Verify identity, continuity, accessibility and educational appropriateness.

Compiler Impact: Produces the Character Engine™ block for the BCube Gold Production Prompt™.

CE-POS-001

Requirement: Poses MUST communicate purposeful educational actions.

Rationale: Maintain recognisable, trustworthy and educationally effective characters.

Inputs: Illustration metadata, curriculum context, page objective and continuity profile.

Outputs: Character specification, pose, expression and interaction metadata.

Validation: Verify identity, continuity, accessibility and educational appropriateness.

Compiler Impact: Produces the Character Engine™ block for the BCube Gold Production Prompt™.

CE-INT-001

Requirement: Character interactions SHALL reinforce the learning objective.

Rationale: Maintain recognisable, trustworthy and educationally effective characters.

Inputs: Illustration metadata, curriculum context, page objective and continuity profile.

Outputs: Character specification, pose, expression and interaction metadata.

Validation: Verify identity, continuity, accessibility and educational appropriateness.

Compiler Impact: Produces the Character Engine™ block for the BCube Gold Production Prompt™.

CE-CON-001

Requirement: Wardrobe, colours and proportions MUST remain continuous across related pages.

Rationale: Maintain recognisable, trustworthy and educationally effective characters.

Inputs: Illustration metadata, curriculum context, page objective and continuity profile.

Outputs: Character specification, pose, expression and interaction metadata.

Validation: Verify identity, continuity, accessibility and educational appropriateness.

Compiler Impact: Produces the Character Engine™ block for the BCube Gold Production Prompt™.

CE-DIV-001

Requirement: Character representation SHALL reflect diversity and inclusion.

Rationale: Maintain recognisable, trustworthy and educationally effective characters.

Inputs: Illustration metadata, curriculum context, page objective and continuity profile.

Outputs: Character specification, pose, expression and interaction metadata.

Validation: Verify identity, continuity, accessibility and educational appropriateness.

Compiler Impact: Produces the Character Engine™ block for the BCube Gold Production Prompt™.

CE-ACC-001

Requirement: Character behaviour MUST model safe, respectful and age-appropriate conduct.

Rationale: Maintain recognisable, trustworthy and educationally effective characters.

Inputs: Illustration metadata, curriculum context, page objective and continuity profile.

Outputs: Character specification, pose, expression and interaction metadata.

Validation: Verify identity, continuity, accessibility and educational appropriateness.

Compiler Impact: Produces the Character Engine™ block for the BCube Gold Production Prompt™.

CE-MTD-001

Requirement: Character metadata SHALL be complete before prompt compilation.

Rationale: Maintain recognisable, trustworthy and educationally effective characters.

Inputs: Illustration metadata, curriculum context, page objective and continuity profile.

Outputs: Character specification, pose, expression and interaction metadata.

Validation: Verify identity, continuity, accessibility and educational appropriateness.

Compiler Impact: Produces the Character Engine™ block for the BCube Gold Production Prompt™.

CE-VAL-001

Requirement: Character validation MUST pass before final QA and Prompt Compiler execution.

Rationale: Maintain recognisable, trustworthy and educationally effective characters.

Inputs: Illustration metadata, curriculum context, page objective and continuity profile.

Outputs: Character specification, pose, expression and interaction metadata.

Validation: Verify identity, continuity, accessibility and educational appropriateness.

Compiler Impact: Produces the Character Engine™ block for the BCube Gold Production Prompt™.

Character Metadata Matrix

Field	Purpose	Required
Character ID	Canonical reference	Yes
Role	Star/Teacher/Child/Parent	Yes
Expression	Emotional state	Yes
Pose	Educational action	Yes
Wardrobe	Continuity	Optional
Interaction	Learning relationship	Yes
Scene Priority	Primary/Secondary	Yes
Accessibility Notes	Inclusive representation	Optional

Character Anti-Patterns

- Inconsistent Star mascot appearance
- Unexplained wardrobe changes
- Fearful or aggressive expressions without educational purpose
- Unsafe behaviour modelling
- Stereotypical representation
- Characters obstructing activity spaces
- Loss of continuity across sequential pages

Cross References

Related rules: IE-CNT-001, VG-FOC-001, QA-CHR-001, PC-COMP-001.

Expansion Roadmap

Future revisions will expand every Character Engine™ rule with pose libraries, expression catalogs, continuity matrices, accessibility guidance, reusable prompt fragments, validation algorithms and reference illustrations.

Phase 9 - Quality Assurance Engine

Phase 9 - Quality Assurance Engine

Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.

Purpose

The Quality Assurance Engine™ provides the mandatory validation framework for every BCube production prompt and generated page. It verifies educational quality, publishing compliance, illustration consistency, accessibility, brand integrity, and production readiness before a page is approved for publication.

1. QA Philosophy

Objective: Validate every production artifact against BCube Gold standards.

- Quality by design
- Child-first validation
- Evidence-based review
- Continuous improvement
- Zero critical defects

Compiler Output

Produces QA reports, validation metadata, Gold Quality Score™, release recommendation and defect log.

Acceptance Criteria

Only prompts and pages meeting mandatory quality thresholds proceed to publication.

2. Multi-Gate Review Model

Objective: Validate every production artifact against BCube Gold standards.

- Gate 1: Educational Review
- Gate 2: Design & Publishing Review
- Gate 3: Illustration Review
- Gate 4: Accessibility Review
- Gate 5: Final Production Approval

Compiler Output

Produces QA reports, validation metadata, Gold Quality Score™, release recommendation and defect log.

Acceptance Criteria

Only prompts and pages meeting mandatory quality thresholds proceed to publication.

3. Validation Categories

Objective: Validate every production artifact against BCube Gold standards.

- Educational alignment
- Visual hierarchy
- Publishing standards
- Character consistency
- Technical print readiness
- Brand compliance

Compiler Output

Produces QA reports, validation metadata, Gold Quality Score™, release recommendation and defect log.

Acceptance Criteria

Only prompts and pages meeting mandatory quality thresholds proceed to publication.

4. Automated Validation Rules

Objective: Validate every production artifact against BCube Gold standards.

- Metadata completeness
- Required fields
- Version compatibility
- Template compliance
- Prompt structure integrity

Compiler Output

Produces QA reports, validation metadata, Gold Quality Score™, release recommendation and defect log.

Acceptance Criteria

Only prompts and pages meeting mandatory quality thresholds proceed to publication.

5. Manual Review Checklist

Objective: Validate every production artifact against BCube Gold standards.

- Learning objective visible
- Teacher interaction present
- Parent extension valid
- Illustration quality
- Whitespace
- Readability

Compiler Output

Produces QA reports, validation metadata, Gold Quality Score™, release recommendation and defect log.

Acceptance Criteria

Only prompts and pages meeting mandatory quality thresholds proceed to publication.

6. Accessibility Validation

Objective: Validate every production artifact against BCube Gold standards.

- Contrast
- Age readability
- Inclusive representation
- Safe learning environments

Compiler Output

Produces QA reports, validation metadata, Gold Quality Score™, release recommendation and defect log.

Acceptance Criteria

Only prompts and pages meeting mandatory quality thresholds proceed to publication.

7. Gold Quality Score™

Objective: Validate every production artifact against BCube Gold standards.

- Educational 25%
- Publishing 20%
- Illustration 20%
- Design 15%
- Accessibility 10%
- Brand 10%

Compiler Output

Produces QA reports, validation metadata, Gold Quality Score™, release recommendation and defect log.

Acceptance Criteria

Only prompts and pages meeting mandatory quality thresholds proceed to publication.

8. Defect Classification

Objective: Validate every production artifact against BCube Gold standards.

- Critical
- Major
- Minor
- Cosmetic
- Documentation

Compiler Output

Produces QA reports, validation metadata, Gold Quality Score™, release recommendation and defect log.

Acceptance Criteria

Only prompts and pages meeting mandatory quality thresholds proceed to publication.

9. Release Decision

Objective: Validate every production artifact against BCube Gold standards.

- Pass
- Conditional Pass
- Rework Required
- Rejected

Compiler Output

Produces QA reports, validation metadata, Gold Quality Score™, release recommendation and defect log.

Acceptance Criteria

Only prompts and pages meeting mandatory quality thresholds proceed to publication.

10. Continuous Improvement

Objective: Validate every production artifact against BCube Gold standards.

- Feedback loop
- Root cause analysis
- Metrics dashboard
- Version updates

Compiler Output

Produces QA reports, validation metadata, Gold Quality Score™, release recommendation and defect log.

Acceptance Criteria

Only prompts and pages meeting mandatory quality thresholds proceed to publication.

Gold Quality Score Matrix

Category	Weight	Minimum Pass
Educational	25%	23/25
Publishing	20%	18/20
Illustration	20%	18/20
Design	15%	13/15
Accessibility	10%	9/10
Brand	10%	9/10

Negative QA Conditions

- Missing learning objective
- Incorrect page geometry
- Unsafe classroom depiction
- Inconsistent Star mascot
- Unreadable text
- Cropped activity area
- Watermarks or unintended text
- Failure to meet minimum Gold score

Phase 9 Deliverables

- Quality Assurance Engine™ specification
- Five-Gate Review Model
- Gold Quality Score™
- Validation Checklist
- Defect Classification Guide
- Release Decision Framework
- Compiler-ready QA Block

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Master_Part_11.docx`

Purpose

The Quality Assurance Engine™ defines mandatory quality gates for every BCube production asset. It governs validation workflows, defect classification, scoring, release readiness, auditability and continuous improvement before any prompt or page is approved.

QA-GAT-001

Requirement: Every production artifact SHALL pass all mandatory quality gates.

Rationale: Ensure consistent educational, visual and production quality.

Inputs: Outputs from all production engines plus compiled metadata.

Outputs: QA status, score, defect log and release recommendation.

Validation: Automated checks followed by human review where required.

Compiler Impact: Blocks or approves final prompt compilation based on QA status.

QA-EDU-001

Requirement: Educational compliance MUST be validated before release.

Rationale: Ensure consistent educational, visual and production quality.

Inputs: Outputs from all production engines plus compiled metadata.

Outputs: QA status, score, defect log and release recommendation.

Validation: Automated checks followed by human review where required.

Compiler Impact: Blocks or approves final prompt compilation based on QA status.

QA-DES-001

Requirement: Design and publishing compliance SHALL be verified.

Rationale: Ensure consistent educational, visual and production quality.

Inputs: Outputs from all production engines plus compiled metadata.

Outputs: QA status, score, defect log and release recommendation.

Validation: Automated checks followed by human review where required.

Compiler Impact: Blocks or approves final prompt compilation based on QA status.

QA-VIS-001

Requirement: Visual grammar and illustration quality MUST meet Gold standards.

Rationale: Ensure consistent educational, visual and production quality.

Inputs: Outputs from all production engines plus compiled metadata.

Outputs: QA status, score, defect log and release recommendation.

Validation: Automated checks followed by human review where required.

Compiler Impact: Blocks or approves final prompt compilation based on QA status.

QA-CHR-001

Requirement: Character consistency SHALL be validated across related pages.

Rationale: Ensure consistent educational, visual and production quality.

Inputs: Outputs from all production engines plus compiled metadata.

Outputs: QA status, score, defect log and release recommendation.

Validation: Automated checks followed by human review where required.

Compiler Impact: Blocks or approves final prompt compilation based on QA status.

QA-ACC-001

Requirement: Accessibility compliance MUST be reviewed.

Rationale: Ensure consistent educational, visual and production quality.

Inputs: Outputs from all production engines plus compiled metadata.

Outputs: QA status, score, defect log and release recommendation.

Validation: Automated checks followed by human review where required.

Compiler Impact: Blocks or approves final prompt compilation based on QA status.

QA-SCR-001

Requirement: Every artifact SHALL receive a Gold Quality Score™.

Rationale: Ensure consistent educational, visual and production quality.

Inputs: Outputs from all production engines plus compiled metadata.

Outputs: QA status, score, defect log and release recommendation.

Validation: Automated checks followed by human review where required.

Compiler Impact: Blocks or approves final prompt compilation based on QA status.

QA-DEF-001

Requirement: Defects SHALL be classified as Critical, Major, Minor or Cosmetic.

Rationale: Ensure consistent educational, visual and production quality.

Inputs: Outputs from all production engines plus compiled metadata.

Outputs: QA status, score, defect log and release recommendation.

Validation: Automated checks followed by human review where required.

Compiler Impact: Blocks or approves final prompt compilation based on QA status.

QA-REL-001

Requirement: Only approved artifacts MAY proceed to publication.

Rationale: Ensure consistent educational, visual and production quality.

Inputs: Outputs from all production engines plus compiled metadata.

Outputs: QA status, score, defect log and release recommendation.

Validation: Automated checks followed by human review where required.

Compiler Impact: Blocks or approves final prompt compilation based on QA status.

QA-AUD-001

Requirement: All validation results SHALL be recorded for audit.

Rationale: Ensure consistent educational, visual and production quality.

Inputs: Outputs from all production engines plus compiled metadata.

Outputs: QA status, score, defect log and release recommendation.

Validation: Automated checks followed by human review where required.

Compiler Impact: Blocks or approves final prompt compilation based on QA status.

Gold Quality Score Matrix

Category	Weight	Pass Threshold
Educational	25%	23/25
Publishing	20%	18/20
Design	15%	13/15
Illustration	20%	18/20
Accessibility	10%	9/10

Category	Weight	Pass Threshold
Brand & Consistency	10%	9/10

Defect Classification

- Critical – prevents release.
- Major – significant quality issue requiring correction.
- Minor – acceptable temporarily but scheduled for correction.
- Cosmetic – aesthetic improvement only.

Cross References

Related rules: PE-VAL-001, DE-VAL-001, VG-VAL-001, EE-VAL-001, TE-VAL-001, PP-VAL-001, IE-VAL-001, CE-VAL-001, PC-VAL-001.

Expansion Roadmap

Future editions will define automated validation algorithms, scoring formulas, audit reports, regression testing, certification workflows and enterprise quality dashboards.

Phase 10 - Prompt Compiler

Phase 10 - Prompt Compiler

Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.

Purpose

The Prompt Compiler™ is the orchestration engine that assembles validated outputs from all previous engines into a single BCube Gold Production Prompt™. It manages inheritance, variable substitution, template selection, version compatibility, overrides, metadata and final compilation before AI generation.

1. Compiler Philosophy

Objective: Produce one authoritative production prompt from standardized engine outputs.

- Single source of truth
- Reusable engine outputs
- Deterministic compilation
- Traceable versions
- Repeatable production

Compiler Output

Produces a fully compiled BCube Gold Production Prompt™, complete metadata, validation record and release-ready prompt package.

Acceptance Criteria

Compilation is deterministic, versioned, reproducible and free of unresolved conflicts.

2. Compilation Pipeline

Objective: Produce one authoritative production prompt from standardized engine outputs.

- Load Publishing Engine
- Load Design Engine
- Load Visual Grammar Engine
- Load Educational Engine
- Load Teaching Engine
- Load Parent Partnership Engine
- Load Illustration Engine
- Load Character Engine
- Load QA Engine
- Merge page metadata
- Generate final prompt

Compiler Output

Produces a fully compiled BCube Gold Production Prompt™, complete metadata, validation record and release-ready prompt package.

Acceptance Criteria

Compilation is deterministic, versioned, reproducible and free of unresolved conflicts.

3. Input Schema

Objective: Produce one authoritative production prompt from standardized engine outputs.

- Book ID
- Series
- Grade
- Page number
- Page title
- Learning objective
- Template ID
- Engine versions
- Overrides
- Assets

Compiler Output

Produces a fully compiled BCube Gold Production Prompt™, complete metadata, validation record and release-ready prompt package.

Acceptance Criteria

Compilation is deterministic, versioned, reproducible and free of unresolved conflicts.

4. Variable Resolution

Objective: Produce one authoritative production prompt from standardized engine outputs.

- Global variables
- Book variables
- Unit variables
- Page variables
- Character variables
- Theme variables
- Localization

Compiler Output

Produces a fully compiled BCube Gold Production Prompt™, complete metadata, validation record and release-ready prompt package.

Acceptance Criteria

Compilation is deterministic, versioned, reproducible and free of unresolved conflicts.

5. Override Rules

Objective: Produce one authoritative production prompt from standardized engine outputs.

- Page overrides

- Book overrides
- Template overrides
- Protected fields
- Conflict resolution

Compiler Output

Produces a fully compiled BCube Gold Production Prompt™, complete metadata, validation record and release-ready prompt package.

Acceptance Criteria

Compilation is deterministic, versioned, reproducible and free of unresolved conflicts.

6. Metadata & Versioning

Objective: Produce one authoritative production prompt from standardized engine outputs.

- Semantic versions
- Compilation timestamp
- Engine signatures
- Prompt ID
- Change history

Compiler Output

Produces a fully compiled BCube Gold Production Prompt™, complete metadata, validation record and release-ready prompt package.

Acceptance Criteria

Compilation is deterministic, versioned, reproducible and free of unresolved conflicts.

7. Validation Before Output

Objective: Produce one authoritative production prompt from standardized engine outputs.

- All engines present
- Schema complete
- No conflicts
- QA passed
- Version compatible

Compiler Output

Produces a fully compiled BCube Gold Production Prompt™, complete metadata, validation record and release-ready prompt package.

Acceptance Criteria

Compilation is deterministic, versioned, reproducible and free of unresolved conflicts.

8. Output Structure

Objective: Produce one authoritative production prompt from standardized engine outputs.

- Technical specification
- Educational specification
- Teaching specification
- Illustration specification
- Validation metadata
- Compiler footer

Compiler Output

Produces a fully compiled BCube Gold Production Prompt™, complete metadata, validation record and release-ready prompt package.

Acceptance Criteria

Compilation is deterministic, versioned, reproducible and free of unresolved conflicts.

9. Error Handling

Objective: Produce one authoritative production prompt from standardized engine outputs.

- Missing module
- Version mismatch
- Schema error
- Protected-field violation
- Compilation failure report

Compiler Output

Produces a fully compiled BCube Gold Production Prompt™, complete metadata, validation record and release-ready prompt package.

Acceptance Criteria

Compilation is deterministic, versioned, reproducible and free of unresolved conflicts.

10. Release & Governance

Objective: Produce one authoritative production prompt from standardized engine outputs.

- Compiler approval
- Immutable compiled prompt
- Audit trail
- Release package
- Continuous improvement

Compiler Output

Produces a fully compiled BCube Gold Production Prompt™, complete metadata, validation record and release-ready prompt package.

Acceptance Criteria

Compilation is deterministic, versioned, reproducible and free of unresolved conflicts.

Compilation Flow

Page Metadata → Engine Loading → Variable Resolution → Override Resolution → Schema Validation → QA Verification → Prompt Assembly → Metadata Injection → Final BCube Gold Production Prompt™

Compiler Metadata Schema

Field	Purpose	Required
Prompt ID	Unique compiled prompt	Yes
Engine Versions	Traceability	Yes
Template ID	Layout reference	Yes
Compilation Date	Audit	Yes
Book Version	Release control	Yes
Language	Localization	Optional
Approval Status	Release gate	Yes

Phase 10 Deliverables

- Prompt Compiler™ specification
- Compilation pipeline
- Variable resolution framework
- Override policy
- Metadata schema
- Release packaging standard
- Compiler validation checklist
- BCube Gold Production Prompt™ assembly rules

Completion Statement

With Phases 1–10 complete, BCube Prompt Engine™ v3.0 establishes a complete production architecture capable of compiling standardized, traceable and reusable production prompts for the BCube publishing ecosystem.

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Master_Part_12.docx`

Purpose

The Prompt Compiler™ is the orchestration layer that assembles validated outputs from every BCube engine into a deterministic, traceable BCube Gold Production Prompt™. It governs compilation order, inheritance, variable resolution, overrides, metadata, version compatibility and release packaging.

PC-LOD-001

Requirement: The compiler SHALL load engine outputs in the approved execution sequence.

Rationale: Ensure repeatable, deterministic prompt generation.

Inputs: Outputs from Publishing, Design, Visual Grammar, Educational, Teaching, Parent, Illustration, Character and QA engines.

Outputs: Compiled BCube Gold Production Prompt™, metadata and audit record.

Validation: Schema validation, dependency checks, conflict detection and version compatibility.

Compiler Impact: Defines final prompt assembly and release behaviour.

PC-SCH-001

Requirement: All required metadata MUST conform to the master schema before compilation.

Rationale: Ensure repeatable, deterministic prompt generation.

Inputs: Outputs from Publishing, Design, Visual Grammar, Educational, Teaching, Parent, Illustration, Character and QA engines.

Outputs: Compiled BCube Gold Production Prompt™, metadata and audit record.

Validation: Schema validation, dependency checks, conflict detection and version compatibility.

Compiler Impact: Defines final prompt assembly and release behaviour.

PC-VAR-001

Requirement: Variables SHALL resolve using global, book, unit and page precedence.

Rationale: Ensure repeatable, deterministic prompt generation.

Inputs: Outputs from Publishing, Design, Visual Grammar, Educational, Teaching, Parent, Illustration, Character and QA engines.

Outputs: Compiled BCube Gold Production Prompt™, metadata and audit record.

Validation: Schema validation, dependency checks, conflict detection and version compatibility.

Compiler Impact: Defines final prompt assembly and release behaviour.

PC-OVR-001

Requirement: Overrides MAY modify only approved overridable fields.

Rationale: Ensure repeatable, deterministic prompt generation.

Inputs: Outputs from Publishing, Design, Visual Grammar, Educational, Teaching, Parent, Illustration, Character and QA engines.

Outputs: Compiled BCube Gold Production Prompt™, metadata and audit record.

Validation: Schema validation, dependency checks, conflict detection and version compatibility.

Compiler Impact: Defines final prompt assembly and release behaviour.

PC-MRG-001

Requirement: Engine outputs SHALL merge without conflicting instructions.

Rationale: Ensure repeatable, deterministic prompt generation.

Inputs: Outputs from Publishing, Design, Visual Grammar, Educational, Teaching, Parent, Illustration, Character and QA engines.

Outputs: Compiled BCube Gold Production Prompt™, metadata and audit record.

Validation: Schema validation, dependency checks, conflict detection and version compatibility.

Compiler Impact: Defines final prompt assembly and release behaviour.

PC-VAL-001

Requirement: Compilation MUST halt on unresolved validation failures.

Rationale: Ensure repeatable, deterministic prompt generation.

Inputs: Outputs from Publishing, Design, Visual Grammar, Educational, Teaching, Parent, Illustration, Character and QA engines.

Outputs: Compiled BCube Gold Production Prompt™, metadata and audit record.

Validation: Schema validation, dependency checks, conflict detection and version compatibility.

Compiler Impact: Defines final prompt assembly and release behaviour.

PC-MTD-001

Requirement: Compiled prompts SHALL include complete versioned metadata.

Rationale: Ensure repeatable, deterministic prompt generation.

Inputs: Outputs from Publishing, Design, Visual Grammar, Educational, Teaching, Parent, Illustration, Character and QA engines.

Outputs: Compiled BCube Gold Production Prompt™, metadata and audit record.

Validation: Schema validation, dependency checks, conflict detection and version compatibility.

Compiler Impact: Defines final prompt assembly and release behaviour.

PC-PKG-001

Requirement: The compiler SHALL generate a release-ready prompt package.

Rationale: Ensure repeatable, deterministic prompt generation.

Inputs: Outputs from Publishing, Design, Visual Grammar, Educational, Teaching, Parent, Illustration, Character and QA engines.

Outputs: Compiled BCube Gold Production Prompt™, metadata and audit record.

Validation: Schema validation, dependency checks, conflict detection and version compatibility.

Compiler Impact: Defines final prompt assembly and release behaviour.

PC-AUD-001

Requirement: Every compilation SHALL produce an auditable execution record.

Rationale: Ensure repeatable, deterministic prompt generation.

Inputs: Outputs from Publishing, Design, Visual Grammar, Educational, Teaching, Parent, Illustration, Character and QA engines.

Outputs: Compiled BCube Gold Production Prompt™, metadata and audit record.

Validation: Schema validation, dependency checks, conflict detection and version compatibility.

Compiler Impact: Defines final prompt assembly and release behaviour.

PC-REL-001

Requirement: Only QA-approved prompts MAY be released for production.

Rationale: Ensure repeatable, deterministic prompt generation.

Inputs: Outputs from Publishing, Design, Visual Grammar, Educational, Teaching, Parent, Illustration, Character and QA engines.

Outputs: Compiled BCube Gold Production Prompt™, metadata and audit record.

Validation: Schema validation, dependency checks, conflict detection and version compatibility.

Compiler Impact: Defines final prompt assembly and release behaviour.

Compilation Sequence

1. 1. Load global configuration
2. 2. Load book metadata
3. 3. Load page metadata
4. 4. Execute engine outputs in defined order
5. 5. Resolve variables
6. 6. Apply approved overrides
7. 7. Validate schema and dependencies
8. 8. Assemble final prompt
9. 9. Inject metadata and version information
10. 10. Produce release package and audit log

Compiler Metadata Schema

Field	Purpose	Required
Prompt ID	Unique identifier	Yes
Engine Versions	Traceability	Yes
Book Version	Release control	Yes
Template ID	Layout reference	Yes
Compilation Timestamp	Audit	Yes

Field	Purpose	Required
Approval Status	Release gate	Yes
Language	Localization	Optional
Checksum	Integrity verification	Optional

Compiler Anti-Patterns

- Skipping engine outputs
- Resolving conflicting instructions silently
- Compiling with failed QA
- Missing metadata
- Unversioned prompt releases
- Manual edits after compilation without revision tracking

Completion Note

With Parts I–XII complete, the first-pass specification covers every core engine of the BCube Prompt Engine™ v3.0. Future expansions will deepen each rule into detailed operational standards, machine-readable schemas, reference implementations and enterprise governance.

Phase 13 - Master Rule Library

Phase 13 - Master Rule Library

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Purpose

The Master Rule Library™ is the canonical registry of all normative rules used by BCube Prompt Engine™. It provides unique identifiers, ownership, dependencies, priorities, inheritance and lifecycle status.

Rule Identifier Standard

- PE-xxx Publishing Engine
- DE-xxx Design Engine
- VG-xxx Visual Grammar Engine
- EE-xxx Educational Engine
- TE-xxx Teaching Engine
- PP-xxx Parent Partnership Engine
- IE-xxx Illustration Engine
- CE-xxx Character Engine
- QA-xxx Quality Assurance Engine
- PC-xxx Prompt Compiler

Canonical Rule Record

Field	Description	Example	Required	Owner	Lifecycle
Rule ID	Unique identifier	PE-GEO-001	Yes	Publishing	Active
Priority	Execution priority	Critical	Yes	Engine	Active
Dependencies	Required upstream rules	DE-GRID-001	No	Compiler	Active
Validation	Verification method	Geometry check	Yes	QA	Active
Version	Introduced version	3.0.0	Yes	Governance	Active
Status	Draft/Active/Deprecated	Active	Yes	Governance	Active

Inheritance Model

- Global rules apply to every prompt.
- Book rules extend global rules.
- Unit rules extend book rules.
- Page rules extend unit rules.

- Overrides require explicit approval and never bypass mandatory safety or QA rules.

Rule Lifecycle

- Draft → Review → Approved → Active → Superseded → Deprecated → Archived
- Every change requires version increment and audit trail.
- Breaking changes require major version updates.

Expansion Roadmap

The complete Master Rule Library will ultimately catalogue thousands of rules with cross-references, machine-readable schemas, examples, validation logic and compiler mappings.

Phase 14 - Prompt Fragment Library

Phase 14 - Prompt Fragment Library

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Complete Production Standard Phase XIV – Prompt Fragment Library™

Purpose

The Prompt Fragment Library™ is the reusable component catalog used by the Prompt Compiler™. Instead of authoring complete prompts from scratch, the compiler assembles validated fragments into a BCube Gold Production Prompt™.

Publishing Fragments

- PF-PUB-001 Page Geometry
- PF-PUB-002 Safe Margins
- PF-PUB-003 Print Specification
- PF-PUB-004 Branding

Design Fragments

- PF-DES-001 Grid System
- PF-DES-002 Typography
- PF-DES-003 Whitespace
- PF-DES-004 Layout Zones

Educational Fragments

- PF-EDU-001 Learning Objective
- PF-EDU-002 Competency Mapping
- PF-EDU-003 Activity Design
- PF-EDU-004 Assessment Evidence

Teaching Fragments

- PF-TEA-001 Teacher Facilitation
- PF-TEA-002 Question Prompts
- PF-TEA-003 Feedback
- PF-TEA-004 Observation

Illustration Fragments

- PF-ILL-001 Rendering Style
- PF-ILL-002 Environment
- PF-ILL-003 Lighting
- PF-ILL-004 Props

Character Fragments

- PF-CHR-001 Star Mascot
- PF-CHR-002 Teacher
- PF-CHR-003 Child
- PF-CHR-004 Parent

Fragment Record Structure

Field	Description	Required	Example	Owner
Fragment ID	Unique reusable block	Yes	PF-ILL-001	Illustration
Purpose	Function of fragment	Yes	Rendering Style	Engine
Inputs	Required metadata	Yes	Page type	Compiler
Outputs	Generated prompt text	Yes	Prompt block	Compiler
Version	Fragment revision	Yes	3.0.0	Governance

Assembly Principles

- Fragments are immutable once approved.
- The Prompt Compiler selects fragments based on metadata.
- Fragments may inherit global and book-level defaults.
- Fragments never override mandatory safety or QA rules.
- Every compiled prompt records fragment versions for traceability.

Expansion Roadmap

Future editions will include hundreds of reusable prompt fragments with parameter definitions, placeholders, localization support, examples, anti-patterns, and machine-readable schemas.

Phase 15 - Template Library

Phase 15 - Template Library

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Complete Production Standard Phase XV (Template Library™)

Purpose

The Template Library™ defines the canonical page structures used across all BCube publications. Templates separate layout from content so the Prompt Compiler™ can combine standardized page architectures with reusable prompt fragments.

Core Workbook Templates

- Cover
- Inside Cover
- Contents
- Unit Opener
- Learning Page
- Activity Page
- Worksheet
- Review
- Assessment
- Certificate
- Back Cover

Teaching Resources

- Lesson Plan
- Teacher Guide
- Observation Record
- Intervention Plan
- Extension Activity

Parent Resources

- Home Learning Sheet
- Parent Guide
- Weekly Summary
- Family Activity Card

Marketing & Brand

- Poster
- Flyer
- Brochure
- Banner

- Social Media Post
- Magazine Page

Digital Templates

- Interactive Activity
- Digital Worksheet
- Presentation Slide
- Dashboard Card

Canonical Template Record

Field	Description	Example	Required	Owner	Versioned
Template ID	Unique identifier	TPL-ACT-001	Yes	Design	Yes
Category	Template family	Activity	Yes	Publishing	Yes
Purpose	Usage	Tracing page	Yes	Curriculum	No
Layout Zones	Protected regions	Header/Activity/Footer	Yes	Design	Yes
Compatible Engines	Required engines	PE+DE+VG	Yes	Compiler	Yes
Status	Lifecycle	Active	Yes	Governance	Yes

Template Design Principles

- Templates SHALL preserve protected learning zones.
- Templates MUST be reusable across books and grades where applicable.
- Templates SHALL support localization and accessibility.
- Templates MUST remain backward compatible within the same major version.
- Template changes require governance approval and version updates.

Expansion Roadmap

Future editions will provide detailed specifications for every template including geometry, component maps, spacing systems, example pages, validation rules, compiler bindings and print-ready layouts.

Phase 16 - Validation Library

Phase 16 - Validation Library

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Complete Production Standard Phase XVI (Validation Library™)

Purpose

The Validation Library™ defines reusable validation rules, automated quality checks, scoring models and compliance criteria applied across every BCube production artifact before release.

Validation Domains

- Publishing Validation
- Design Validation
- Visual Grammar Validation
- Educational Validation
- Teaching Validation
- Parent Validation
- Illustration Validation
- Character Validation
- Prompt Compiler Validation

Validation Types

- Schema validation
- Metadata validation
- Rule compliance
- Accessibility validation
- Print readiness
- Educational alignment
- Consistency checks
- Regression validation

Execution Levels

- Real-time author validation
- Pre-compilation validation
- Compilation validation
- Pre-release validation
- Post-release audit

Canonical Validation Rule

Field	Description	Example	Required	Severity	Auto
VAL-ID	Unique validation ID	VAL-PUB-001	Yes	Critical	Yes

Field	Description	Example	Required	Severity	Auto
Engine	Owning engine	Publishing	Yes	Major	Yes
Rule	Requirement tested	Safe margin	Yes	Critical	Yes
Method	Validation approach	Geometry check	Yes	Major	Yes
Failure Action	System response	Block release	Yes	Critical	Yes
Evidence	Audit artifact	Validation report	Optional	Minor	No

Gold Quality Thresholds

- Critical validations: 100% pass required.
- Major validations: ≥98% pass required.
- Minor validations: ≥95% pass required.
- Cosmetic issues tracked but do not block release unless policy requires.

Validation Pipeline

1. Author validation
2. Engine validation
3. Cross-engine validation
4. Compiler validation
5. QA certification
6. Release approval
7. Audit archival

Expansion Roadmap

Future editions will define thousands of validation rules, scoring algorithms, automated test suites, reference datasets, dashboards and certification workflows.

Phase 17 - BCube Gold Certification Standard

Phase 17 - BCube Gold Certification Standard

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Complete Production Standard Phase XVII (BCube Gold Certification Standard™)

Purpose

The BCube Gold Certification Standard™ defines the formal certification program used to determine whether any BCube publication, prompt, illustration, assessment or digital learning asset meets the organization's highest production, educational and quality benchmarks.

Certification Principles

- Certification is evidence-based and repeatable.
- Every certified asset is fully traceable.
- Certification requires successful completion of all mandatory validation gates.
- Certification status is version-specific.
- All certification decisions are auditable.

Certification Levels

Level	Description	Minimum Score	Release
Bronze	Basic compliance	85%	Internal
Silver	High quality	92%	Pilot
Gold	Commercial production	97%	Approved
Platinum	Reference standard	99%+	Flagship

Mandatory Certification Gates

- Educational compliance
- Publishing compliance
- Design compliance
- Visual Grammar compliance
- Teaching compliance
- Parent Partnership compliance
- Illustration compliance
- Character compliance
- Accessibility compliance
- Prompt Compiler compliance
- Quality Assurance approval

Evidence Requirements

- Validation reports
- Gold Quality Score™
- Version manifest
- Rule compliance report
- Visual review checklist
- Educational review record
- Audit log
- Release approval

Certification Lifecycle

1. Submission
2. Automated validation
3. Expert review
4. Certification decision
5. Publication
6. Surveillance review
7. Recertification after major revisions

Revocation Conditions

- Critical defect discovered after release
- Failure to maintain mandatory standards
- Unauthorized modification
- Version inconsistency
- Audit failure

Expansion Roadmap

Future editions will define assessor qualifications, certification workflows, sampling plans, compliance dashboards, statistical quality metrics, renewal schedules and enterprise governance policies.

Phase 18 - Enterprise Governance and Version Control

Phase 18 - Enterprise Governance and Version Control

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Complete Production Standard Phase XVIII (Enterprise Governance & Version Control™)

Purpose

Enterprise Governance & Version Control™ establishes the management framework for the BCube Prompt Engine™. It defines ownership, approval authority, change control, release management, document lifecycle, compliance responsibilities and long-term maintenance of the production standard.

Governance Principles

- Single authoritative source for all production standards.
- Every change is traceable and version controlled.
- All normative rules require documented approval.
- Governance decisions are evidence-based and auditable.
- Standards evolve through controlled releases.

Governance Roles

Role	Primary Responsibility	Approval Authority	Examples
Product Owner	Vision and roadmap	Final	Founder / Product Lead
Curriculum Board	Educational standards	Educational	Academic reviewers
Design Board	Visual standards	Design	Creative leads
QA Board	Quality & certification	Release	QA leadership
Governance Office	Version control	Policy	Documentation team

Versioning Policy

- Major versions introduce breaking architectural changes (e.g. 3.0 → 4.0).
- Minor versions add backward-compatible capabilities (e.g. 3.1).
- Patch versions correct defects without changing behavior (e.g. 3.1.1).
- Every release includes a changelog, approval record and compatibility statement.

Change Management Workflow

1. Proposal submitted
2. Impact assessment
3. Technical review

4. Educational review
5. Design review
6. Quality review
7. Approval
8. Implementation
9. Regression validation
10. Release publication
11. Archive previous version

Document Lifecycle

- Draft
- Internal Review
- Approved
- Released
- Maintained
- Superseded
- Archived

Governance Metrics

- Standards compliance rate
- Validation pass rate
- Certification success rate
- Average review cycle time
- Rule reuse percentage
- Template reuse percentage
- Defect escape rate

Completion Statement

Phase XVIII completes the foundational governance framework for BCube Prompt Engine™ v3.0. Future work will expand the standard with detailed rule catalogs, reference implementations, prompt libraries, automated tooling specifications and enterprise operating procedures.

Normative expansion from `BCube_Prompt_Engine_v3_Complete_Production_Standard_Phase_18_Enterprise_Governance_and_Version_Control.docx`

Complete Production Standard Phase XVIII (Enterprise Governance & Version Control™)

Purpose

Enterprise Governance & Version Control™ establishes the governance framework for the BCube Prompt Engine™. It defines ownership, decision rights, change management, release governance, auditability, documentation lifecycle and long-term maintenance.

Governance Structure

Role	Primary Responsibility
Founder / Chief Standards Officer	Owns strategic direction and approves major standards.
Editorial Standards Board	Maintains educational and publishing standards.
Technical Standards Committee	Owns compiler, schemas and automation.
Quality Review Board	Approves Gold Certification decisions.
Release Manager	Coordinates production releases and version control.

Versioning Policy

- Semantic versioning SHALL be used (Major.Minor.Patch).
- Major versions indicate architectural or breaking changes.
- Minor versions add backward-compatible capabilities.
- Patch versions correct defects without changing behavior.
- Every release SHALL include release notes and migration guidance.

Change Management Workflow

1. Proposal
2. Impact Assessment
3. Technical Review
4. Editorial Review
5. Pilot Validation
6. Approval
7. Release
8. Post-release Review

Document Lifecycle

- Draft → Internal Review → Approved → Published → Maintained → Superseded → Archived
- Every revision SHALL retain an audit trail.
- Deprecated rules remain searchable until formally archived.

Compliance & Audit

- Annual standards review
- Quarterly rule audit
- Release compliance verification
- Random publication sampling
- Corrective action tracking

Future Enterprise Roadmap

Future editions will define governance charters, approval matrices, RACI models, digital workflow integration, automated policy enforcement, API specifications, analytics dashboards and enterprise operating metrics.

Phase 19 - Reference Implementation and Example Library

Phase 19 - Reference Implementation and Example Library

> *Normative status: This module is part of the BCube Prompt Engine v3.0 Complete Production Standard. MUST, SHALL, SHOULD, MAY, and MUST NOT carry their formal specification meanings.*

Complete Production Standard Phase XIX (Reference Implementation & Example Library™)

Purpose

The Reference Implementation & Example Library™ provides canonical examples, reference prompts, annotated page specifications, anti-patterns and implementation guidance. It demonstrates how all BCube engines work together in real production scenarios.

Reference Assets

- Workbook page examples
- Cover examples
- Assessment examples
- Teacher guide examples
- Parent guide examples
- Digital lesson examples

Example Types

- Canonical example
- Minimal example
- Advanced example
- Localization example
- Accessibility example
- Edge-case example

Anti-Pattern Library

- Layout failures
- Educational misalignment
- Visual clutter
- Prompt conflicts
- Metadata omissions
- QA failures

Implementation Guides

- Prompt assembly walkthrough
- Engine interaction maps
- Compiler trace examples
- Validation reports
- Release checklist

Canonical Example Record

Field	Description	Example	Required	Validated	Versioned
Example ID	Unique identifier	EX-ACT-001	Yes	Yes	Yes
Template	Associated template	Activity Page	Yes	Yes	Yes
Prompt	Compiled prompt	Gold Prompt	Yes	Yes	Yes
Expected Output	Reference result	Illustrated page	Yes	Yes	No
Validation	QA evidence	Gold Certified	Yes	Yes	Yes
Notes	Implementation guidance	Best practice	Optional	No	Yes

Reference Library Principles

- Every reference example SHALL be fully traceable.
- Examples MUST be reproducible from the published standard.
- Anti-patterns SHALL explain why an implementation fails.
- Reference outputs SHOULD include QA evidence and compiler metadata.
- All examples are version-controlled.

Expansion Roadmap

Future editions will include hundreds of fully annotated reference pages, complete prompt pairs, before-and-after comparisons, automated regression examples, multilingual examples, and implementation packs for every BCube publication type.